

Custom Compiler™ Pre-2020.03 Interactive Router User Guide

Version T-2022.06, June 2022



Copyright and Proprietary Information Notice

© 2022 Synopsys, Inc. This Synopsys software and all associated documentation are proprietary to Synopsys, Inc. and may only be used pursuant to the terms and conditions of a written license agreement with Synopsys, Inc. All other use, reproduction, modification, or distribution of the Synopsys software or the associated documentation is strictly prohibited.

Destination Control Statement

All technical data contained in this publication is subject to the export control laws of the United States of America. Disclosure to nationals of other countries contrary to United States law is prohibited. It is the reader's responsibility to determine the applicable regulations and to comply with them.

Disclaimer

SYNOPSYS, INC., AND ITS LICENSORS MAKE NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

Trademarks

Synopsys and certain Synopsys product names are trademarks of Synopsys, as set forth at <https://www.synopsys.com/company/legal/trademarks-brands.html>. All other product or company names may be trademarks of their respective owners.

Free and Open-Source Licensing Notices

If applicable, Free and Open-Source Software (FOSS) licensing notices are available in the product installation.

Third-Party Links

Any links to third-party websites included in this document are for your convenience only. Synopsys does not endorse and is not responsible for such websites and their practices, including privacy practices, availability, and content.

www.synopsys.com

Contents

New in This Release	5
Related Products, Publications, and Trademarks	5
Conventions	5
Customer Support	6
Statement on Inclusivity and Diversity	7
1. Interactive Routing	8
About the Pre-2020.03 Interactive Router	9
Enabling the Pre-2020.03 Interactive Router	9
Checking for Design Errors	10
About the Preroute Checker	10
Design Errors	11
Design Warnings	13
Technology Errors	15
Running the Preroute Checker	15
Viewing and Resolving Design Errors	17
Routing Using the Follow-the-Cursor Method	17
Routing Using the Point-to-Point Method	20
Adjusting Route Widths Automatically	21
Adding Vias Automatically	23
Connecting to Neighboring Devices Automatically	25
Cloning Routes Automatically	29
Adding Vias and Making Route Adjustments	33
Pausing and Making Route Adjustments	34
Add Via Dialog Box Options	36
Rotating Vias	38
Managing Option Sets	39
Opening the Create Route Dialog Box	39
Creating Option Sets	40
Modifying Option Sets	41
Renaming Option Sets	42

Contents

Deleting Option Sets	42
Saving Option Sets to a Directory	42
Reloading Options Sets from a Directory	43
Route Toolbar and Create Route Dialog Box Reference	43
Basic Options	43
General Tab	46
Layer Tab	47
Via Tab	51
Cloning/Welding Tab	53
Shield Tab	55
Tandem Tab	59
2. Interactive Routing Cursor Movement and Behavior	61
Basic Interactive Cursor Movements	61
Interactions with Obstructions	63
Routing Over and Beyond Objects	65
Routing Around Obstacles in Trackless Designs	66
Adding Points Along the Route	67
Adding Vias	69
Unroutable Objects	70
Jogs and Via Costs in Routing	70
Self-Intersecting Routing Patterns and Same-Net Obstructions	74
3. Troubleshooting Error Messages	76
Where the Error Messages Are Displayed	76
Types of Error Messages	78
IGCR_CREATE_ROUTE Error Messages	79

About This User Guide

This user guide explains how to use the Pre-2020.03 Interactive Router to add routes to integrated circuit designs.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the Custom Compiler Release Notes on the SolvNetPlus site.

Related Products, Publications, and Trademarks

For additional information about the Custom Compiler tool, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

You might also want to see the documentation for the following related Synopsys products:

Synopsys® Custom Compiler™

Synopsys Fusion Compiler™

Synopsys IC Compiler™ II

Synopsys IC Compiler™

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
Courier	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>
Courier bold	Indicates user input—text you type verbatim—in examples, such as <code>prompt> write_file top</code>
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN</code> .
	Indicates a choice among alternatives, such as <code>low medium high</code>
\	Indicates a continuation of a command line.
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

Customer Support

Customer support is available through SolvNetPlus.

Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNetPlus site, go to the following address:

<https://solvnetplus.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

Contacting Customer Support

To contact Customer Support, go to <https://solvnetplus.synopsys.com>.

Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

Interactive Routing

This document describes how to use the Pre-2020.03 Interactive Router, the legacy tool that is superseded by the new Interactive Router introduced in the Q-2020.03 release.

This section contains the following topics:

- [About the Pre-2020.03 Interactive Router](#)
- [Enabling the Pre-2020.03 Interactive Router](#)
- [Checking for Design Errors](#)
- [Routing Using the Follow-the-Cursor Method](#)
- [Routing Using the Point-to-Point Method](#)
- [Adjusting Route Widths Automatically](#)
- [Adding Vias Automatically](#)
- [Connecting to Neighboring Devices Automatically](#)
- [Cloning Routes Automatically](#)
- [Adding Vias and Making Route Adjustments](#)
- [Managing Option Sets](#)
- [Route Toolbar and Create Route Dialog Box Reference](#)

For information about the overall routing workflow, how to prepare for routing, setting up routing constraints, other types of routing, and routing error messages, see the *Custom Router User Guide*.

About the Pre-2020.03 Interactive Router

The Pre-2020.03 Interactive Router lets you create routes for single nets interactively. You manually determine the route using one of the following methods, and then the router creates the interconnects automatically:

- **Follow the Cursor:** The router creates the interconnect automatically between points as you click on the canvas.
- **Point to Point (P2P):** The router creates the interconnect automatically after you specify a start point and an end point on the canvas.

During the routing operation, the Pre-2020.03 Interactive Router:

- Automatically creates DRC-aware routes.
- Automatically routes around obstructions.
- Automatically inserts vias (in multi-layer mode).
- Supports contactless metal layers for local interconnect routing.
- Supports automatic tapering.
- Supports design-specific constraints.
- Supports non-default rules (NDRs).
- Automatically snaps individual routes and differential-pair routes to active tracks if you turn on this option.
- (Interactive point-to-point routing only) Automatically snaps parallel shielding and jumpers for parallel shielding to active tracks if you turn on the option for track adherence.

When you use the Pre-2020.03 Interactive Router, consider the following:

- The Pre-2020.03 Interactive Router supports orthogonal routing only.
- Point-to-Point routing supports shielding and differential pair mixed-signal constraints, but does not support other mixed-signal design constraints.

Enabling the Pre-2020.03 Interactive Router

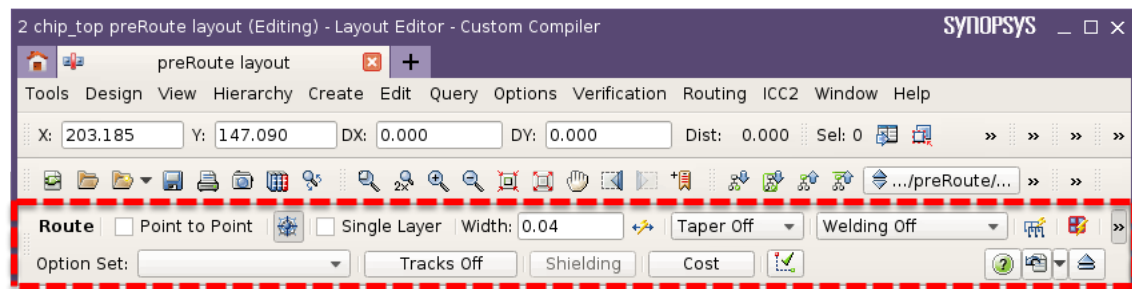
Caution:

Before you start, make sure you review the information in Preparing for Routing. If you want to route a subsection (area) of the design, define the area. For instructions, see Defining a Routing Area.

You must enable the Pre-2020.03 Interactive Router so that you can set the routing options and start the routing operation.

To enable the Pre-2020.03 Interactive Router:

1. In the Layout Editor, make sure the Custom Router is enabled. For instructions, see [Getting Started](#). Make sure the `rtUse202003IR` Tcl preference is `true`.
2. Choose **Routing > Interactive Route**. In the background, the Pre-2020.03 Interactive Router initializes and loads the data. When the initialization and data loading are finished, the Route toolbar appears.



Checking for Design Errors

This topic describes the following:

- [About the Preroute Checker](#)
- [Running the Preroute Checker](#)
- [Viewing and Resolving Design Errors](#)

About the Preroute Checker

Before you start the Pre-2020.03 Interactive Router, you can run the Preroute Checker to make sure the design is free of design-level errors. Doing so ensures that the Interactive Router will run successfully.

The Preroute Checker is a Custom Router feature that checks the design for the following:

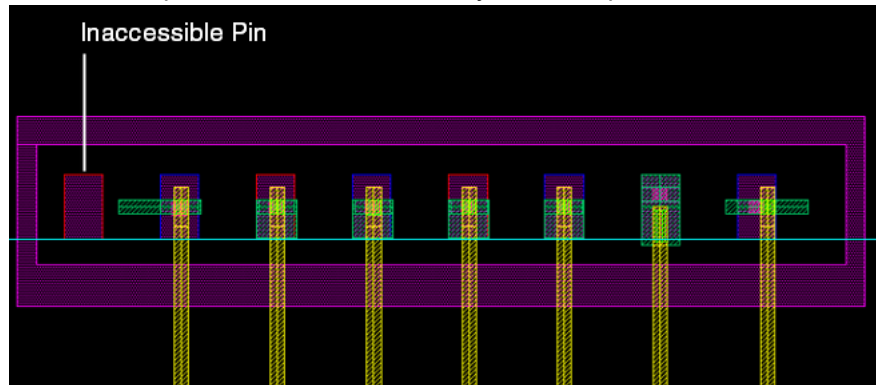
- [Design Errors](#)
- [Design Warnings](#)
- [Technology Errors](#)

Error markers are added in the design and listed in the Error Viewer so that you can easily locate and resolve the errors. In addition, a summary report is displayed in the Console and log file. See [Viewing and Resolving Design Errors](#).

Design Errors

The Custom Router will display error messages for the following design errors:

Type of Design Error	Description
Pin Accessibility	Pins that are not accessibility during routing: • Pins that are not accessible through vias • Pins that are completely blocked • Pin widths that are not within the <code>allowedWidthRange</code> Layer constraint In the following example, the Preroute Checker will display an error for the left-most pin because it is blocked by other shapes.



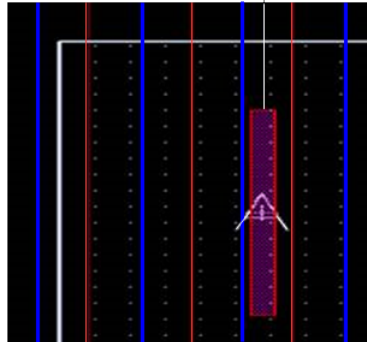
Type of Design Error	Description
----------------------	-------------

Off Track Pin

(Designs that have routing tracks) Pins that are not on routing tracks or have widths different from the tracks.

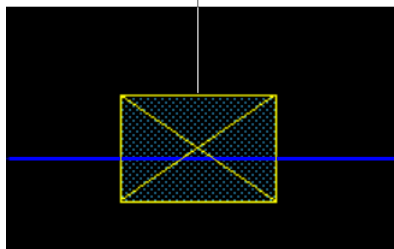
In the following example, the Preroute Checker will display an error for the pin that is placed between the blue and red track.

Off-Track Pin



In the following example, the Preroute Checker will display an error for the pin that is on the blue track, but the center of the pin is not on the track.

Off-Track Pin

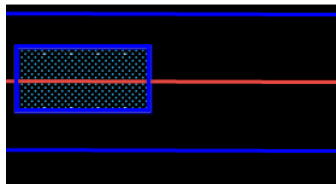


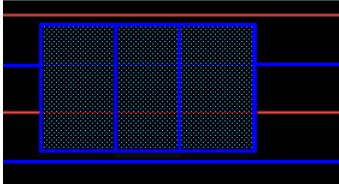
Pin and Track Color

(Designs that have colored routing tracks) Pin colors that do not match track colors.

In the following example, the Preroute Checker will display an error for the blue pin that is placed on the red track.

Blue Pin on Red Track

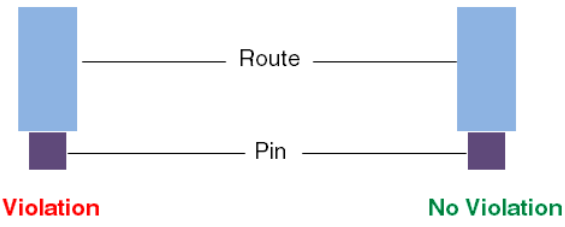
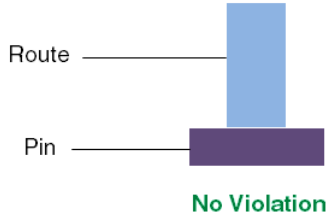


Type of Design Error	Description
Pin Shorts	Short circuits caused by preroute pin shapes that are positioned incorrectly. In the following example, the Preroute Checker will display an error because the two overlapping blue pins are causing short circuitry. Short - Overlapping Pins 

Design Warnings

The Custom Router will display warning messages for the following:

Type of Design Issue	Description
Single Pin in Routing Area	Pins that require connections to other pins that are outside of the defined routing area.
Track Region and Layer Direction Mismatch	(Designs that have routing tracks) The tracks in the track region do not match the routing layer direction. For example, the Preroute Checker will display an error message if the tracks in a track region are vertical, but the direction specified for the routing layer is horizontal.

Type of Design Issue	Description
Pin Size	Pins that have widths that are: - Not within <code>allowedWidthRange</code> and might result in reroutes. For example, the pin width is 0.12, and <code>allowedWidthRange</code> is 0.03 to 0.10. - Narrower than the route layer width constraint, and <code>rtTaperUpAllowed</code> is false. Pin Width < Route Layer Width <code>rtTaperUpAllowed = false</code> <code>rtTaperUpAllowed = true</code>  Pin Width > Route Layer Width <code>rtTaperUpAllowed = true or false</code> 
Unnetted Shape	Shapes that do not have net assignments.
Label on Shapes Without Pins	Shapes that have labels, and the label text should match net names in the design, but: - The shapes do not have pin or net associations. - The net names do not match the label text.
Pin Off Grid Points	(Designs that have routing tracks) Both ends of pins are not on grid points. If only one end of the pin is on the grid point, the Pre-2020.03 Interactive Router will extend the other end to prevent violation.
Nets Without Shapes	Nets that do not have any shapes associated with them.

Technology Errors

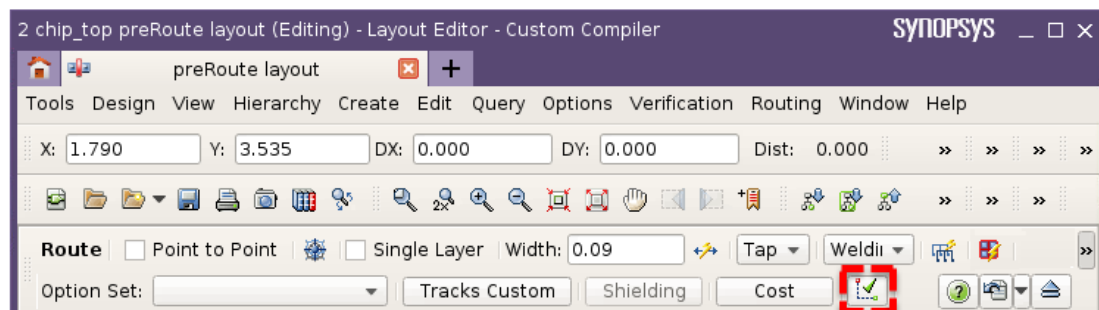
The Custom Router will display error messages for the following errors:

Type of Design Issue	Description
Missing Via Definitions	Missing via definitions between metal layers, starting from M0.
Missing Width Constraints	Missing <code>minWidth</code> constraint for metal, contactless, and poly layers.
Missing Spacing Constraints	Missing spacing constraint for metal, contactless, and poly layers.

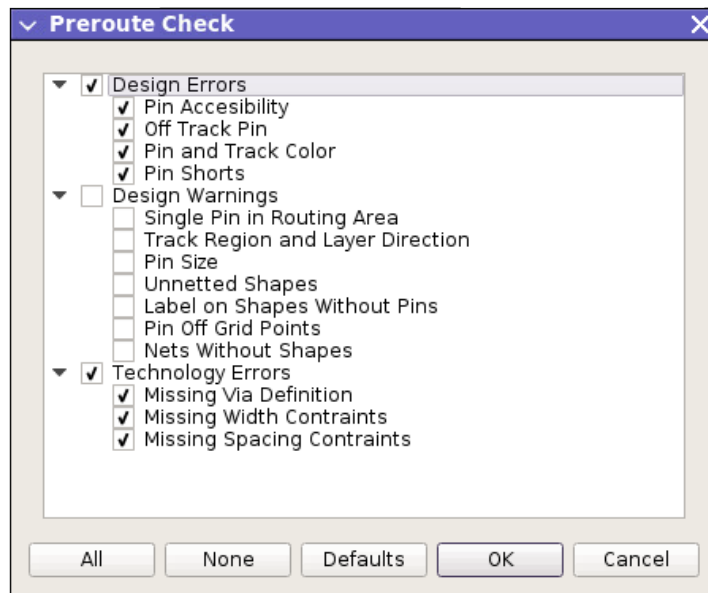
Running the Preroute Checker

To run the Preroute Checker:

1. In the Design Navigator, select the nets you want to check. Alternatively, you can define the area in which you will run the checker.
2. On the **Route** toolbar, click **Preroute Check**.

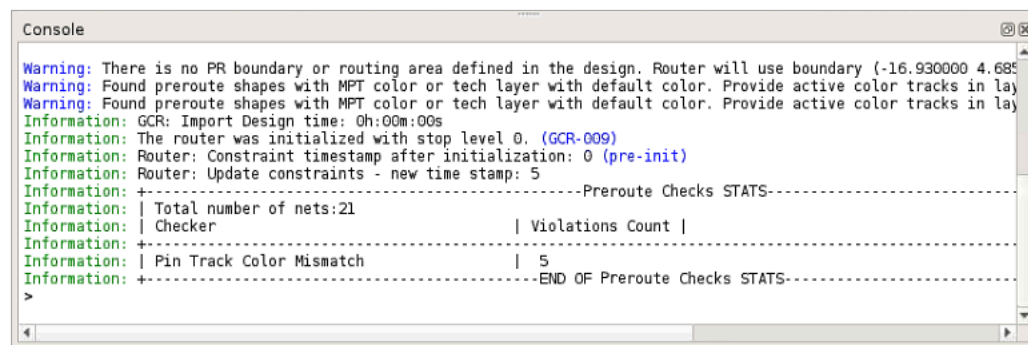


3. In the **Preroute Check** dialog box, select the checks you want to run, and click **OK**.



4. Click **OK**. The Preroute Checker starts checking the design.

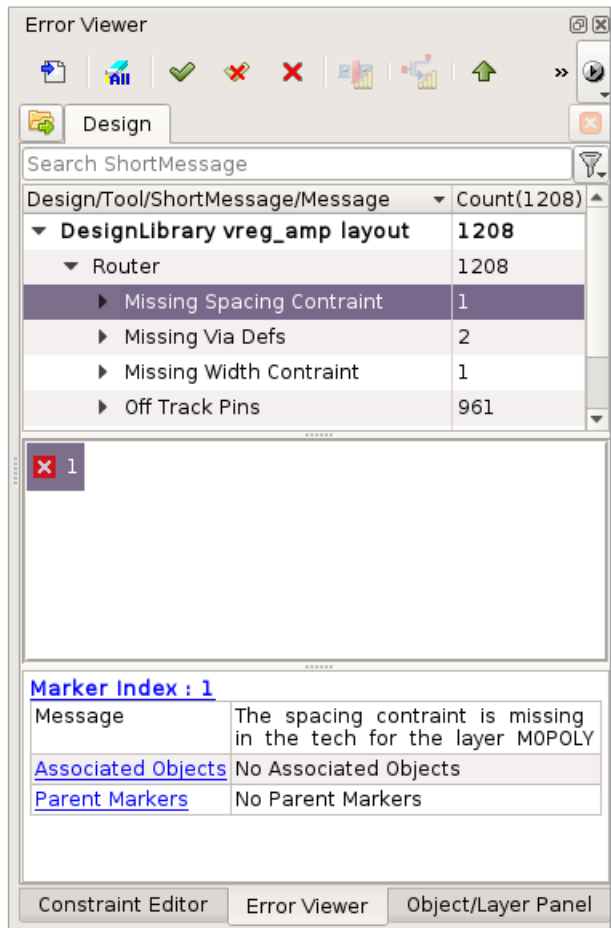
When the checker is finished, error markers appear on the canvas. A summary is available in the Console and log file.



Viewing and Resolving Design Errors

To view and resolve the errors:

1. Display the Error Viewer. The list of errors appears in the viewer.



2. Select an error in the Error Viewer. The viewport zooms in to the location of the error.
3. Resolve the error.

The Error Viewer explains the causes of the errors. See also [Troubleshooting Error Messages](#).

Routing Using the Follow-the-Cursor Method

 [Show me](#) an example of follow-the-cursor routing.

To create a route using the follow-the-cursor method:

1. On the **Route** toolbar, clear the **Point to Point** check box and set the routing options. For information about the **Route** toolbar options, see [Route Toolbar and Create Route Dialog Box Reference](#).
2. Select or clear the **Single Layer** check box, depending on the net that needs to be routed.
3. To enter the first point, click the shape or pin on the net that you want to route. Toggle the active layer (by pressing the 1 key) to ensure that the route begins on the desired layer.
4. Continue clicking to add route points. As you add a new route point, the router adds an interconnect between the last two points.

You should add route points along the desired vertical and horizontal tracks, but stay away from the route bends.

If you make a mistake, you can:

- Use the Backspace key to remove previously entered points.
 - Press Shift+Esc to reset the command.
5. To complete the route:
 - a. Do one of the following:
 - Click the final compatible pin or wire.
 - Double-click any place in the design, including open space.
 - b. Middle-click the last shape or point and choose one of the following:
 - **Commit:** Saves the created route to the database.
 - **Finish Route:** (Last shape only) Routes the interconnect to the nearest compatible pin.
 6. Press Esc to exit the follow-the-cursor routing command.

You can view and edit the route shape properties in the Property Editor. See [Viewing and Modifying Route Attributes](#). For instructions on viewing and editing via shape properties, see [Modifying Via Attributes and Properties](#).

If error messages appear in the Console during routing, see [Troubleshooting Error Messages](#).

The following is an example of follow-the-cursor interactive routing flow:

1. Open your design in the Layout Editor.
2. Use the Constraint Editor to add constraints.
3. Enable the Connectivity Engine and turn on Flightlines for the nets that you want to route. See [Enabling the Connectivity Engine and Displaying and Hiding Flight Lines](#).
4. Turn on the DRD (design-rule driven) tool. See [Checking for Violations as You Edit \(DRD\)](#).
5. (Optional) Specify whether the tool displays resistance, length, and capacitance values on the cursor as you create the route.
6. Initialize and start the router by choosing the **Routing > Interactive Router** command.
7. Select or clear the **Single Layer** check box, depending on the net that needs to be routed.
8. Click on the starting shape.

The tool taps on the net and LPP and creates the route in real time (follow-the-cursor mode).

9. Move the cursor in the direction towards the target figure.

When you do this, the flightline to the closest target is shown on the canvas.

10. Complete the route operation by:
 - Single-clicking when the cursor is above a compatible pin or wire.
 - Double-clicking at any point in the design, including open space.
11. Use the Marker Browser to see if there were any DRC violations created by the Pre-2020.03 Interactive Router.
12. Run DRC on the design, and then modify routes if needed.

Use the Layout Editor to manually edit the design as needed.
13. Run LVS on the design.
14. Save and close the design.

Routing Using the Point-to-Point Method

To create a route using the point-to-point method:

1. On the **Route** toolbar, select **Point to Point** and set the routing options. For information about the **Route** toolbar options, see [Route Toolbar and Create Route Dialog Box Reference](#).
2. Select or clear the **Single Layer** check box, depending on the net that needs to be routed.
3. To enter a point, click the shape or pin on the net that you want to route. Toggle the active layer (by pressing 1) to ensure that the route begins on the desired layer.
4. Continue clicking to specify intermediate guide points.

If you make a mistake, you can:

- Use the Backspace key to remove previously entered points.
 - Press Shift+Esc to reset the command.
5. To complete the route, click a compatible pin or wire.
 6. Press Esc to end the command.

Note:

If the route fails, review the console messages. It is possible that more precise P2P guidance will realize the route.

If error messages appear in the Console during routing, see [Troubleshooting Error Messages](#).

The following is an example of point-to-point interactive routing flow:

1. Open your design in the Layout Editor.
2. Use the Constraint Editor to add constraints (NDRs) on the design.
3. Enable the Connectivity Engine and turn on Flightlines for the nets that you want to route. See [Enabling the Connectivity Engine and Displaying and Hiding Flight Lines](#).
4. Turn on the DRD (design-rule driven) tool. See [Checking for Violations as You Edit \(DRD\)](#).
5. Initialize and start the router by choosing the **Routing > Interactive Router** command.
6. Select **Multi-Layer** or **Single-Layer** depending on the net that needs to be routed.

7. Click on the starting shape.

The tool taps on the net and LPP and creates the route in real time.

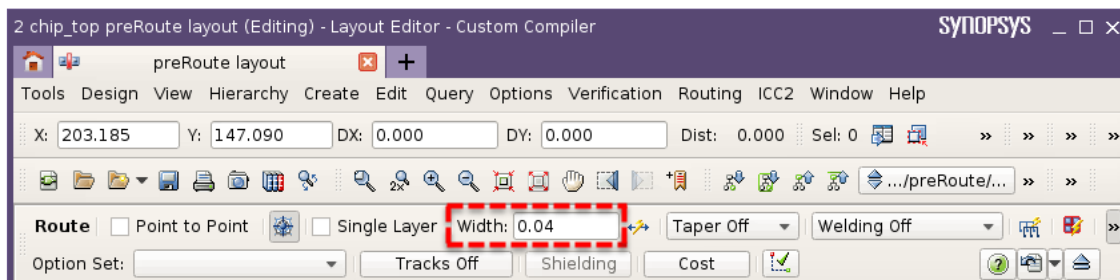
8. Complete the route operation by double-clicking on the target figure.
9. Use the Marker browser to see if there were any DRC violations created by the Pre-2020.03 Interactive Router.
10. Run DRC on the design, and then modify routes if needed.

Use the Layout Editor to manually edit the design as needed.

11. Run LVS on the design.
12. Save and close the design.

Adjusting Route Widths Automatically

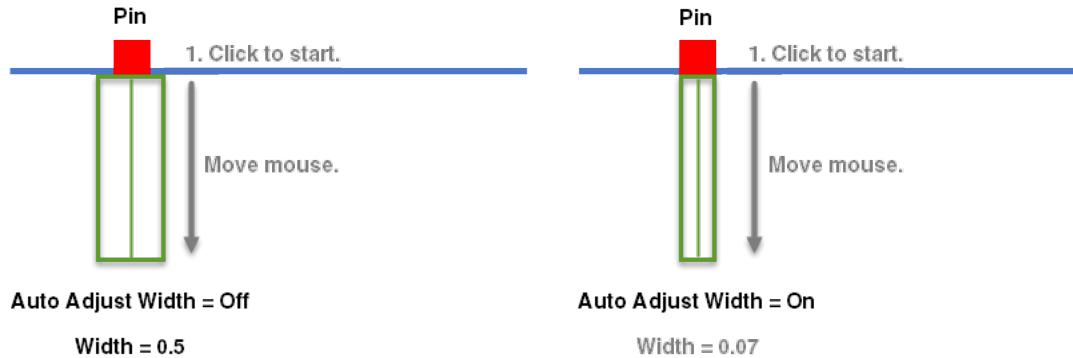
Width and **Auto Adjust Width** are options that you can specify on the **Route** toolbar for follow-the-cursor and point-to-point interactive routing. For the description of the options, see [Route Toolbar and Create Route Dialog Box Reference](#).



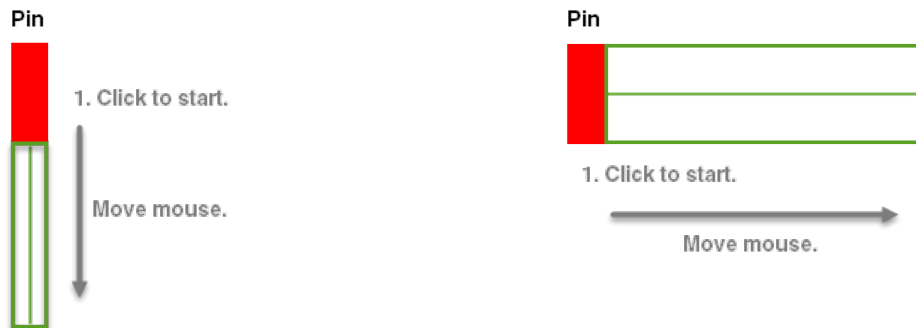
You can use **Width** and  **Auto Adjust Width** together when creating routes. When you start routing, **Auto Adjust Width** automatically applies the width that matches the starting pin geometry.

In the following example, the pin width is 0.07 in both scenarios.

- In the first scenario (left), **Auto Adjust Width** is turned off, and the **Width** value is set to 0.5. When you click the pin to start routing, the first route segment has a width of 0.5, which is wider than the pin width (0.07).
- In the second scenario (right), **Auto Adjust Width** is turned on. The router automatically adjusts the route width to match that of the pin. The **Width** box becomes unavailable for editing and displays the width value applied: 0.07.



When **Auto Adjust Width** is turned on, the width of the first route segment is determined by the dimension of the pin and the direction of the route, as shown in the following examples.



To override the automatically adjusted starting width, you can press Ctrl+4 to cycle through pre-determined width settings. For information about the Ctrl+4 width settings, see [Route Toolbar and Create Route Dialog Box Reference](#).

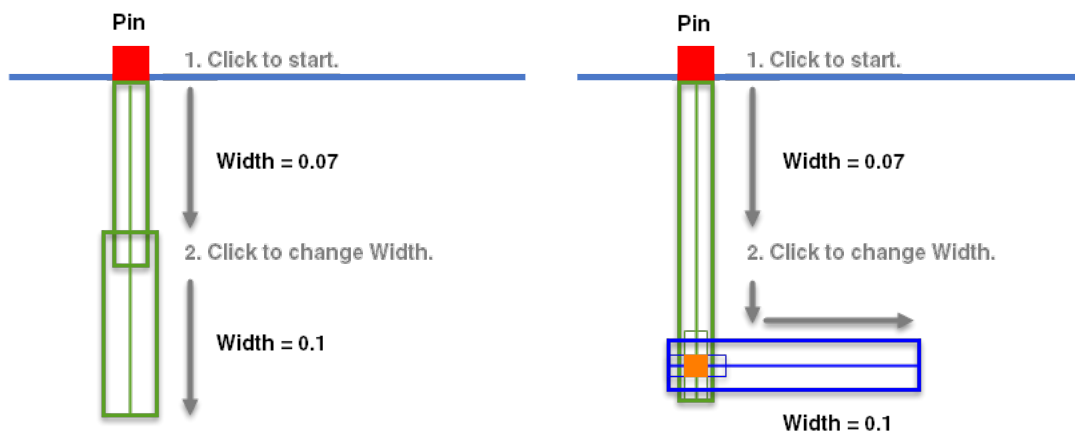
After the first route segment is placed, you can change the route width, regardless of the **Auto Adjust Width** setting. For example, you might want to widen the route to reduce resistance.

To change the route width, click the route at the location where you want the change to occur, and then type the new value in the **Width** box. Alternatively, you can press Ctrl+4 to cycle through the pre-determined width settings.

The new route width is applied as follows:

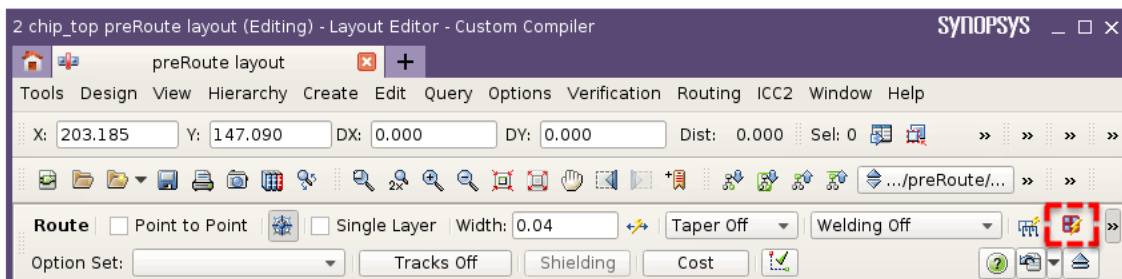
- If you continue the route in the same direction, the new width is applied from the most recently committed point (click) on the current layer, as shown in the first scenario (left).
- If you change the route direction, the new width is applied when you start the new route direction, as shown in the second scenario (right).

If you are in the **Single Layer** mode, the perpendicular route continues on the current layer. If you are in the **Multi Layer** mode, the perpendicular route continues on the adjacent layer, as shown in the second scenario (right).



Adding Vias Automatically

Auto Via Drop is an option that you can select on the **Route** toolbar for follow-the-cursor and point-to-point interactive routing.



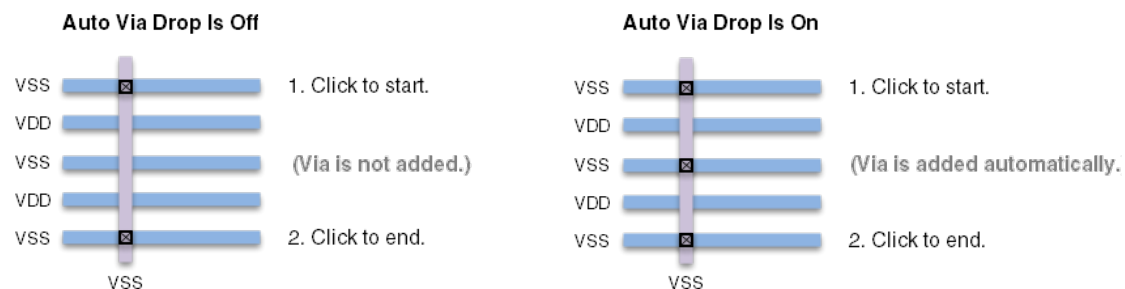
The **Auto Via Drop** can assist you with power grid and device-level routing by automatically adding vias along routes to connect to same-net wires or pins that are one layer above or below.

During Auto Via Drop operations:

- The router will not add stack vias.
- The router will add vias to connect to wires, pre-wires, or pins.
- The router will only add vias if the vias can be fully enclosed by the connecting wires or pins.
- The vias can be single-cut vias or via arrays. Single vias include cut class specified vias.
- Multiple vias can be added along the path being interactively routed.

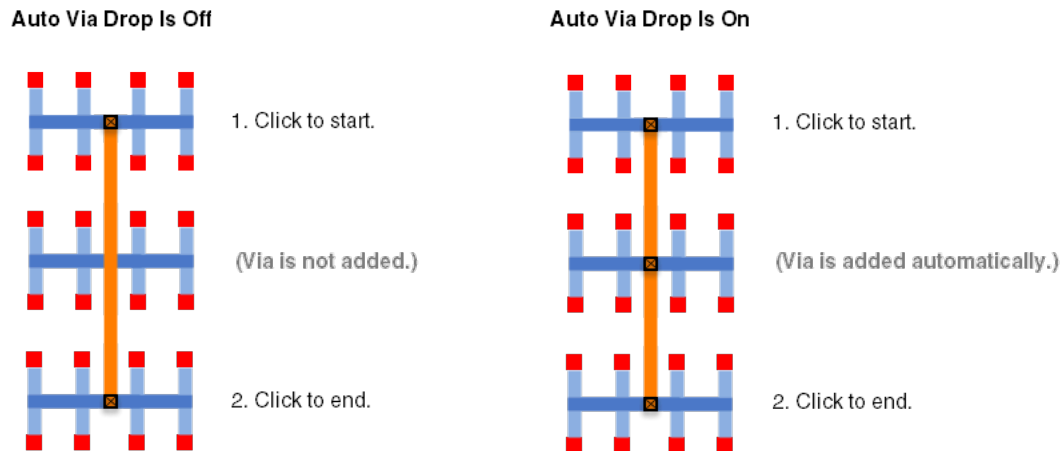
The following power grid examples show the difference between routing without **Auto Via Drop** and routing with **Auto Via Drop**.

- **Auto Via Drop is turned off:** When you create routes in a power grid, vias are added only when you right-click along the route and use the **Via Up**, **Via Down**, and **Add Via** commands. In the example, vias are added manually at the two locations shown.
- **Auto Via Drop is turned on:** As you create routes in the power grid, vias are automatically added to connect to the wire or pin that is on the layer above or below. In the example, vias are automatically added at every VSS-to-VSS and VDD-to-VDD connection as the routes are created across the grid.



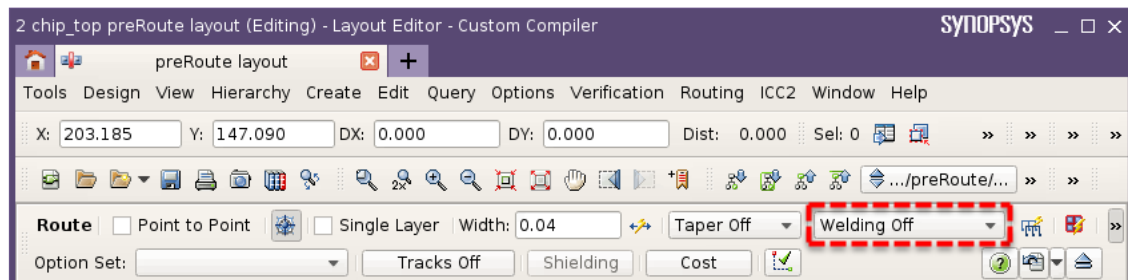
The following auto welding examples show the difference between routing without **Auto Via Drop** and routing with **Auto Via Drop**. For information about auto welding, see [Connecting to Neighboring Devices Automatically](#).

- **Auto Via Drop is turned off:** When you route manually to connect the fishbone structures shown, vias are added only at the starting and ending locations.
- **Auto Via Drop is turned on:** When you route manually to connect the fishbone structures shown, a via is automatically added to connect to the wire (or trunk) in the second fishbone structure.



Connecting to Neighboring Devices Automatically

Auto Welding Mode is an option that you can select on the **Route** toolbar for follow-the-cursor and point-to-point interactive routing.

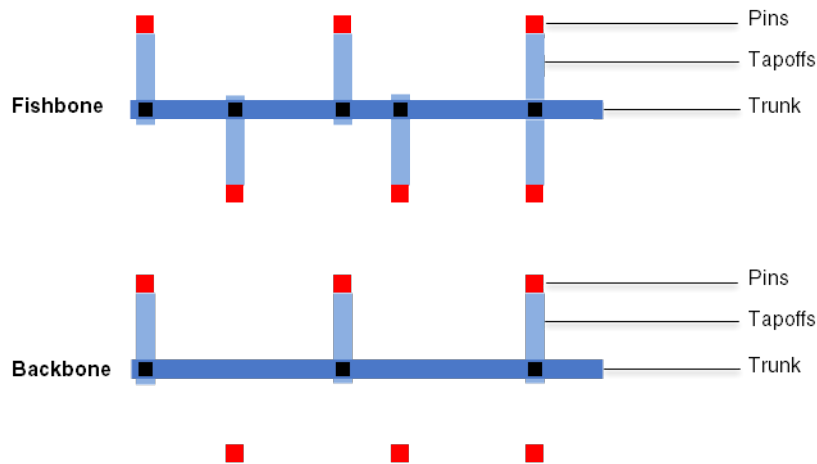


The **Auto Welding Mode** can assist you with power-grid and device-level routing by automatically connecting to pins in neighboring devices or routes. **Auto Welding Mode** is typically used with **Auto Cloning** and **Auto Via Drop**.

During interactive routing, Auto Welding automatically branches (taps off) from the main interconnect (trunk) to connect to nearby pins on the same net in the **Single Layer** or **Multi Layer** modes. Vias are added automatically at the pins. If you select the **Auto Via Drop** option, vias are also added along wires and at pins that are one layer above or below the welding wire. For more information, see [Adding Vias Automatically](#).

You can choose one of the following Auto Welding modes:

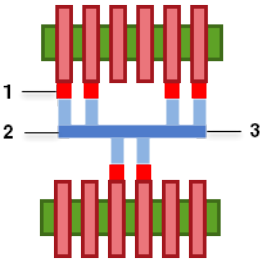
- **Fishbone:** Welding connections occur on both sides of the trunk.
- **Backbone:** Welding connections occur on one side of the trunk only. The side is determined by the side of the starting pin or the first nearest pin the interconnect encounters.



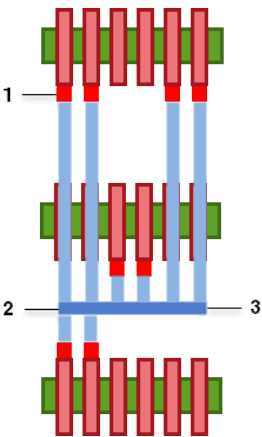
Before you start routing with an **Auto Welding Mode** option, consider the following:

- Synopsys recommends that you use the **Auto Welding Mode** while performing follow-the-cursor interactive routing. Follow-the-cursor routing lets you preview potential welding routes and make adjustments as you move the cursor on the canvas, before you commit the route. Because point-to-point routing does not offer the same level of preview, you would have to reroute to make any changes.
- The router can weld across multiple rows of devices in the entire design. If you want to confine the welding possibilities, define a routing area to restrict the welding operations. For information about defining a routing area, see [Defining a Routing Area](#).
- Note that the connecting pins on the same net do not have to be the same width.
- Welding routes can start at a pin, from any location on a trunk, or from an existing route.
- The maximum number of welding connections is determined by the `rtMaxWeldingConnections` Tcl preference. The default setting is 256.

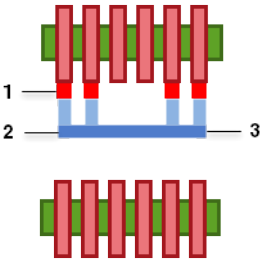
In the following **Fishbone Auto Welding Mode** example, two devices are placed as shown. All of the red pins are on the same net. To route the red pins, you would:

Step	
	1 Click the first pin to start follow-the-cursor interactive routing.
	2 Route downward, and then click midway across the channel to change the route direction. As the route continues horizontally through the channel, the router automatically welds to nearby pins that are on the same net.
	3 Below the last pin, press Enter or double-click to finish the route.

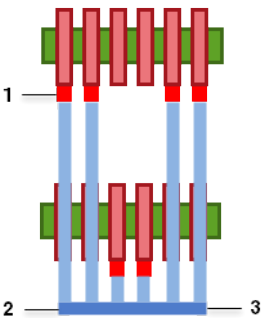
In the following **Fishbone Auto Welding Mode** example, two devices are placed as shown. All of the red pins are on the same net, but they are distributed across two adjacent channels. To route the red pins, you would:

Step	
	1 Click the first pin to start follow-the-cursor interactive routing.
	2 Route downward, and then click midway across the second channel to change the route direction. As the route continues horizontally through the channel, the router automatically welds to nearby pins that are on the same net.
	3 Below the last pin, press Enter or double-click to finish the route.

In the following **Backbone Auto Welding Mode** example, two devices are placed as shown. All of the red pins are on the same net. To route the red pins, you would:

Step	
	1 Click the first pin to start follow-the-cursor interactive routing.
	2 Route downward, and then click midway across the channel to change the route direction. As the route continues horizontally through the channel, the router automatically welds to nearby pins that are on the same net.
	3 Below the last pin, press Enter or double-click to finish the route.

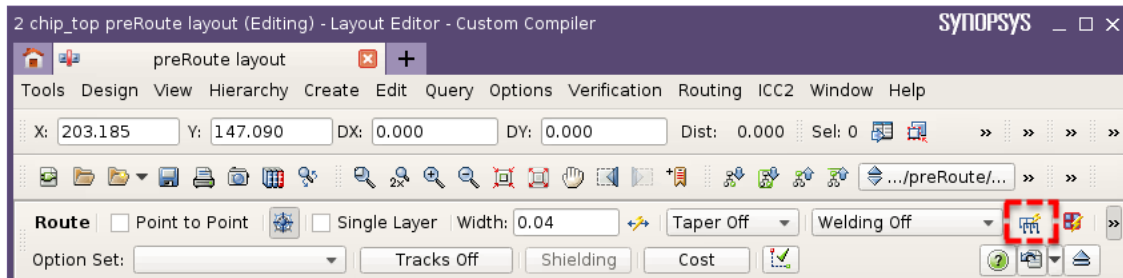
In the following **Backbone Auto Welding Mode** example, two devices are placed as shown. All of the red pins are on the same net, but they are distributed across different channels. To route the red pins, you would:

Step	
	1 Click the first pin to start follow-the-cursor interactive routing.
	2 Route downward, and then click midway across the second channel to change the route direction. As the route continues horizontally through the channel, the router automatically welds to nearby pins on the same net.
	3 Below the last pin, press Enter or double-click to finish the route.

Instead of using the **Auto Welding Mode** options on the **Route** toolbar, you can use the `rtAutoWeldingMode` Tcl preference. For information, see the *Custom Compiler Tcl Reference Manual*.

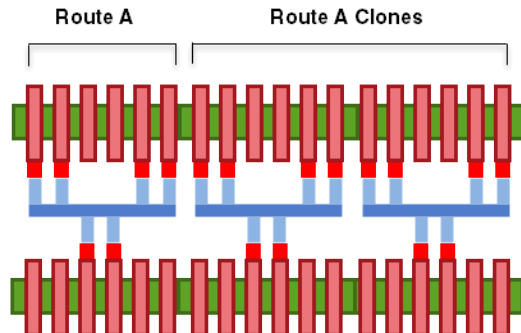
Cloning Routes Automatically

Auto Cloning is an option that you can select on the **Route** toolbar for follow-the-cursor and point-to-point interactive routing.



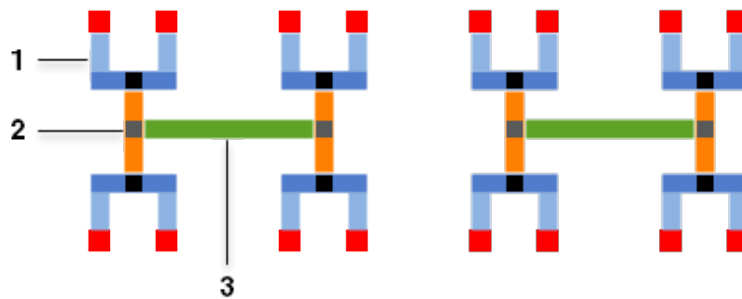
Auto Cloning can assist you with device-level routing by automatically replicating routes within and between rows of devices. **Auto Cloning** is typically used with **Auto Welding Mode** and **Auto Via Drop**.

During interactive routing, the router looks for matching patterns of devices and pins on the same net. If it finds device and pin patterns that match, the router will replicate the route for those devices. In the following example, two clones are created within the same row.



Vias are added automatically at the pins. If you select the **Auto Via Drop** option, vias are also added along wires that are one layer above or below the welding wire. For more information, see [Adding Vias Automatically](#).

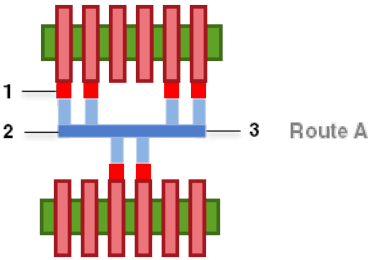
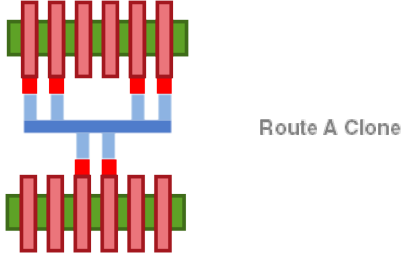
You can use **Auto Cloning** to connect a hierarchy of cloned routes. In the following example, as you connect the first pair of blue routes (1) to create a *spine* (2), **Auto Cloning** copies that spine three times across the channel. Then, as you connect first two orange spines (3), **Auto Cloning** copies the green spine connection in the adjacent matching structure.



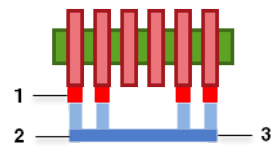
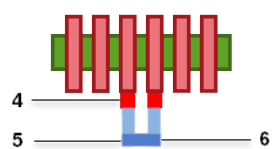
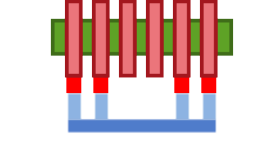
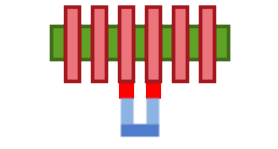
Before you start interactive routing with the **Auto Cloning** option, consider the following:

- Synopsys recommends that you use the **Auto Cloning** option while performing follow-the-cursor interactive routing. Follow-the-cursor routing lets you preview potential welding routes and make adjustments as you move the cursor on the canvas, before you commit the route. Because point-to-point routing does not offer the same level of preview, you would have to reroute to make any changes.
- If you want to confine the cloning operations within a smaller region in the design, define a routing area first. For information about defining a routing area, see [Defining a Routing Area](#).
- The router will not clone existing routes. Cloning can only occur when you create new routes.
- Cloning routes can start at a pin, from any location on a trunk, or from an existing route.
- The maximum number of clones per route is determined by the `rtMaxCloning` Tcl preference. The default setting is 64.

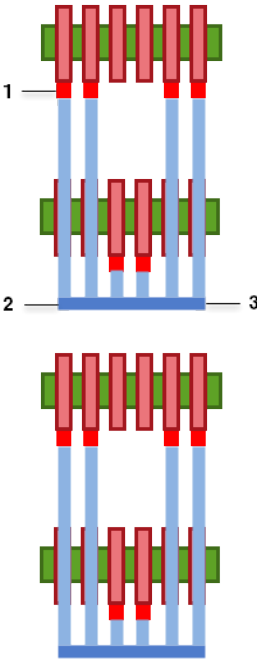
In the following **Auto Cloning** example, four devices are placed as shown, and the **Fishbone Auto Welding Mode** is chosen. All of the red pins are on the same net. To route the red pins, you would:

Step	
	1 Click the first pin to start follow-the-cursor interactive routing.
	2 Route downward, and then click midway across the first channel to change the route direction. As the route continues horizontally through the channel, the router automatically welds to nearby pins on the same net. Because the third and fourth devices are identical to the first two devices and have the same pin positions, the router clones the current route in the third channel.
	3 Below the last pin, press Enter or double-click to finish the route. The cloned route also changes the route direction and finishes at the last pin.

In the following **Auto Cloning** example, four devices are placed as shown, and the **Backbone Auto Welding Mode** is chosen. All of the red pins are on the same net. To route the red pins, you would:

	Step
	1, 4 Click the first pin to start follow-the-cursor interactive routing.
	2, 5 Route downward, and then click midway across the channel to change the route direction. As the route continues horizontally through the channel, the router automatically welds to nearby pins on the same net. Because the third and fourth devices are identical to the first two devices and have the same pin positions, the router clones the current routes in the third and fourth channels.
	3, 6 Below the last pin, press Enter or double-click to finish the route. The cloned routes also change the route direction and finish at the last pin.
	

In the following **Auto Cloning** example, four devices are placed as shown, and the **Backbone Auto Welding Mode** is chosen. All of the red pins are on the same net. To route the red pins, you would:

Step	
	1 Click the first pin to start follow-the-cursor interactive routing.
	2 Route downward, and then click midway across the second channel to change the route direction. As the route continues horizontally through the channel, the router automatically welds to nearby pins on the same net. Because there are matching rows of devices and connections, the router clones the current route and connections as shown.
	3 Below the last pin, press Enter or double-click to finish the route.

Instead of using the **Auto Cloning** option on the **Route** toolbar, you can use the `rtAutoCloning` Tcl preference. For information, see the *Custom Compiler Tcl Reference Manual*.

Adding Vias and Making Route Adjustments

After you start interactive routing, you can pause and resume, add and remove points along the route, add and remove vias, move to an adjacent layer, and make other adjustments.

This topic describes the following:

- [Pausing and Making Route Adjustments](#)
- [Add Via Dialog Box Options](#)
- [Rotating Vias](#)

Pausing and Making Route Adjustments

To pause or make adjustments to routes:

1. At the location where you want to pause or make adjustments, right-click on the canvas. One of the following menus appears.

Interactive Route Method	Menu
--------------------------	------

Follow the Cursor

Pause/Resume FTC	P
Finish Route	Ctrl+Return
Commit	Enter
Add Point	Space
Remove Point	Backspace
☒ Single Layer	
Via Up	Shift+V
Via Down	Ctrl+V
Add Via	V
Rotate Via	R
Options	▶

Single Layer Mode:

Pause/Resume FTC	P
Finish Route	Ctrl+Return
Commit	Enter
Add Point	Space
Remove Point	Backspace
☐ Single Layer	
Layer Up	Shift+V
Layer Down	Ctrl+V
Options	▶

Multi Layer Mode:

Point to Point Mode

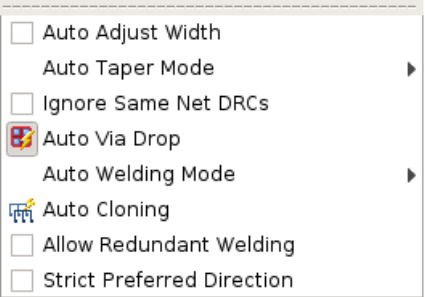
The menu for the Single Layer and Multi Layer modes are

Pause/Resume FTC	P
Finish Route	Ctrl+Return
Commit	Enter
Add Point	Space
Remove Point	Backspace
☐ Single Layer	
Options	▶

identical.

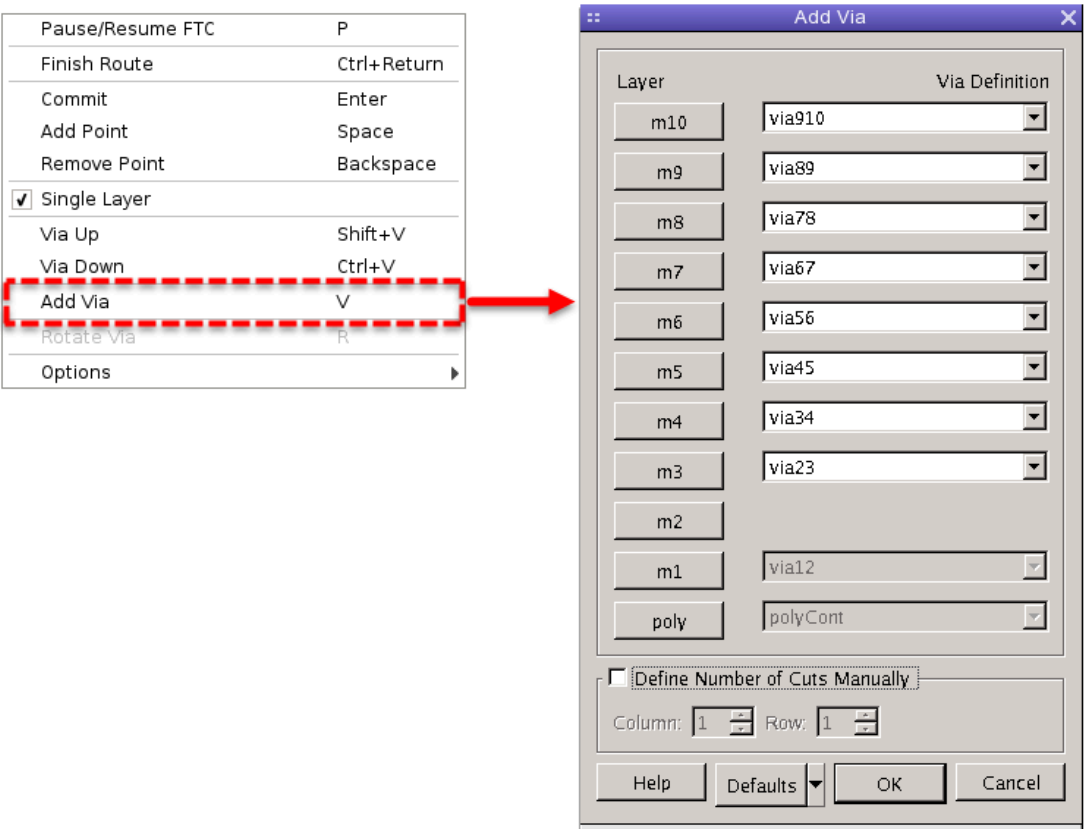
2. Choose one of the following commands or options:

Command	Description	Default Shortcut Keys
Pause/Resume FTC	Pauses or resumes the follow-the-cursor interactive routing.	P
Finish Route	Completes the route to the nearest connection point.	Ctrl+Enter
Commit	Completes the path if it is valid and writes to the database.	Enter
Add Point	Add an intermediate point (follow-the-cursor mode) or guide point (point-to-point mode).	Space
Remove Point	Deletes the previously entered point. In P2P mode, result in the “unfreezing” of the route from the second to last add point event to the last add point event.	Backspace
Single Layer	Switches between the single-layer and multi-layer modes. - Clear the check box: Specifies the multi-layer mode, in which generated physical interconnect can span multiple layers. - Select Single Layer: Specifies the single-layer mode, in which generated physical interconnect can reside on only one layer.	
Layer Up (follow-the-cursor, Multi Layer mode)	Continues routing on metal at the next upper layer.	Shift+V
Layer Down (follow-the-cursor, Multi Layer mode)	Continues routing on metal at the next lower layer.	Ctrl+V
Via Up (follow-the-cursor, Single-Layer mode)	Adds a via above the current layer. See Adding Vias .	Shift+V
Via Down (follow-the-cursor, Single-Layer mode)	Adds a via below the current layer. See Adding Vias .	Ctrl+V

Command	Description	Default Shortcut Keys
Add Via (follow-the-cursor, Single-Layer mode)	Lets you select a specific standard via for a given layer transition. For more information, see Add Via Dialog Box Options. - If you are in an open area of your design, the default via configuration is aligned to the corresponding routing layers preferred routing direction. - If you are not in an open area of the design, the router automatically rotates the pad directions as needed.	V
Rotate Via (follow-the-cursor, Single-Layer mode)	Lets you manually rotate the via that you are adding. See Rotating Vias .	R
Options	Lets you select the desired routing options. The options are also available in the Create Route dialog box. See Route Toolbar and Create Route Dialog Box Reference for details.  The screenshot shows a dialog box with the following options: ☐ Auto Adjust Width - Auto Taper Mode ▶ ☐ Ignore Same Net DRCs ☒ Auto Via Drop - Auto Welding Mode ▶ ☒ Auto Cloning ☐ Allow Redundant Welding ☐ Strict Preferred Direction	

Add Via Dialog Box Options

While using the Pre-2020.03 Interactive Router, you can add vias along the route if you are using the Follow the Cursor - Single Layer mode. For instructions, see [Adding Vias and Making Route Adjustments](#). When you select to add a via along the route, the **Add Via** dialog box opens.



The following table describes the options in the **Add Via** dialog box:

Option	Description
Layer	The possible layers the router can reach from the current follow-the-cursor routing layer. To insert a via to a new routing layer, the user must select, by clicking on the corresponding button, a different layer than the current one. If the layer above or below the current layer is not a valid routing layer, the layer will not be displayed in the interface.
Via Definition	By default, the default via per adjacent layer will be chosen. Users may choose a different standard Via Definition for each layer to move to. Modifications to the Via Definition value will be remembered in the current design session for subsequent Add Via actions but will not apply to Via Up/Down actions which will continue to use the defined standard default via.
Define Number of Cuts Manually	Specifies how to calculate the number of via rows and columns. When disabled, the router calculates number based on the current routing. When enabled, you can manually enter the number of via rows and columns.

Option	Description
Cols/Rows	Defines the number of cut rows and cut columns that the router applies to the parameterized via currently selected to be inserted on the cursor.

Rotating Vias

During Follow-the-Cursor Single-Layer interactive routing, you can right-click the canvas and add vias. See [Pausing and Making Route Adjustments](#).

By default, the vias are aligned with the layer's preferred routing direction. However, you can manually rotate vias in 90° increments. The number of possible configurations depends on whether the rotation involves single-cut vias or multiple-cut vias.

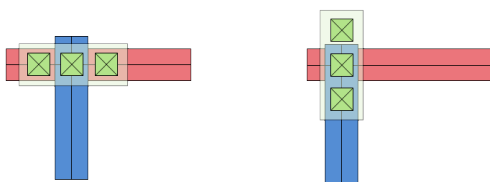
When rotating single-cut vias, four configurations are possible. You can rotate the upper layer 90° by pressing Ctrl+3. Alternatively, you can rotate the lower layer 90° by pressing Ctrl+Shift+3.

Single-cut via rotation configurations:



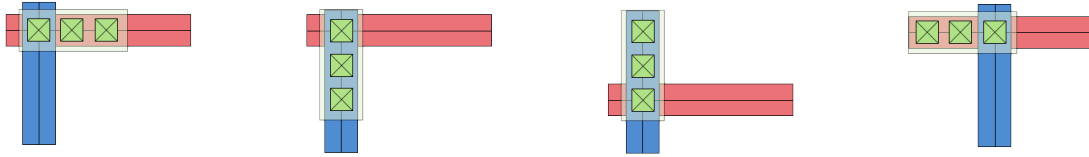
When rotating multiple-cut vias, six configurations are possible. The following examples show two configurations. The center of the via aligns with the intersecting center lines of the wires above and below.

Multiple-cut via rotation configuration when the via is centered:



In the remaining four configurations, the intersecting center lines of the wires align with the multiple-cut via at four different locations: centers of the left-most, top-most, bottom-most, and right-most via. In these configurations, the vias are flagged to the east, south, north, and west, respectively.

Multiple-cut via rotation configurations when the via is flagged:



Managing Option Sets

After you set the options on the **Route** toolbar or **Create Route** dialog box, you can create option sets and recall the settings later within the same design session. Doing so allows you to experiment with different settings without losing them.

You can also save the options to a directory, permitting them to be shared across design sessions or with other users.

This topic describes the following:

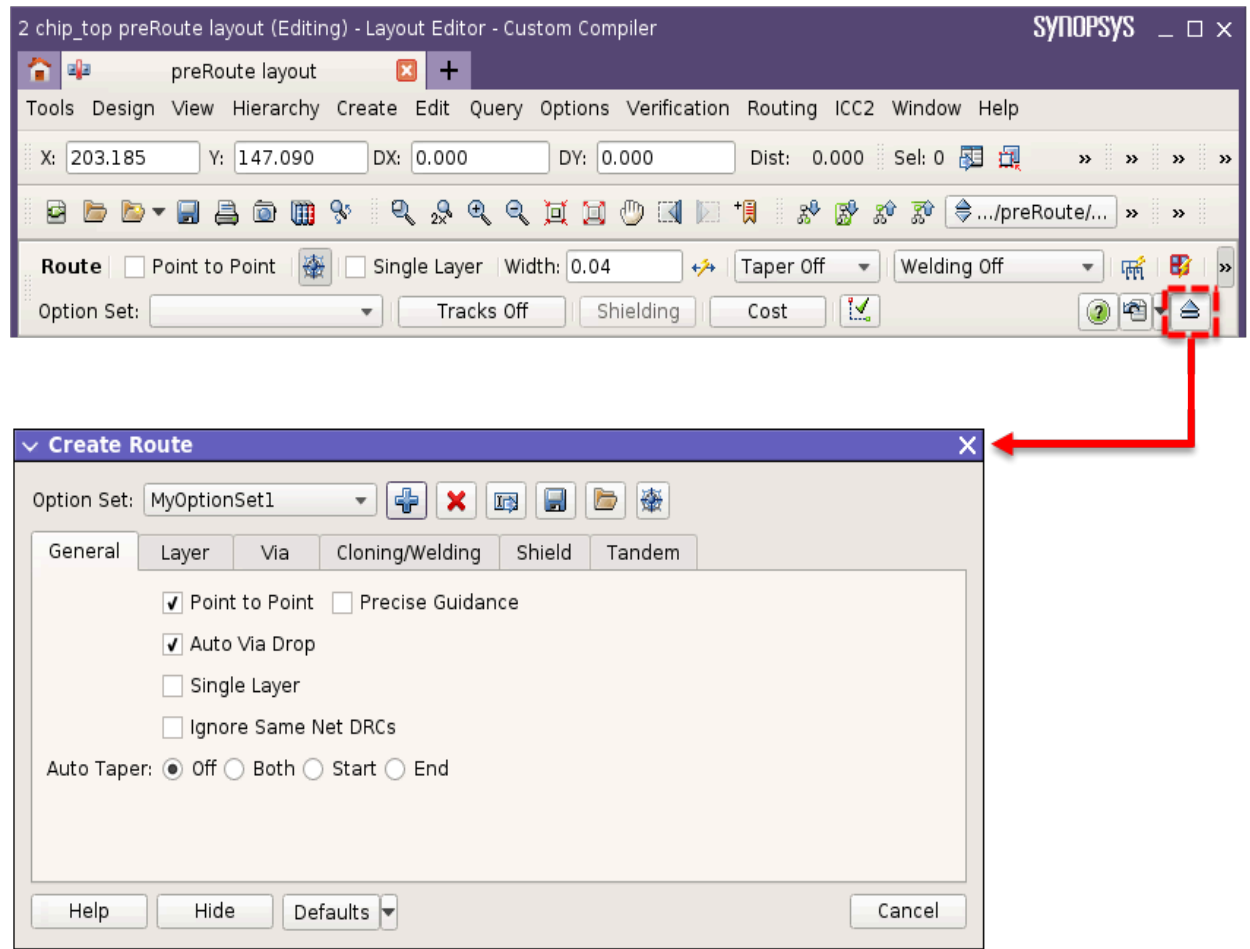
- [Opening the Create Route Dialog Box](#)
- [Creating Option Sets](#)
- [Modifying Option Sets](#)
- [Renaming Option Sets](#)
- [Deleting Option Sets](#)
- [Saving Option Sets to a Directory](#)
- [Reloading Options Sets from a Directory](#)

Opening the Create Route Dialog Box

You create, modify, and save interactive routing options in the **Create Route** dialog box.

To open the **Create Route** dialog box:

- On the **Route** toolbar, click **Eject** .

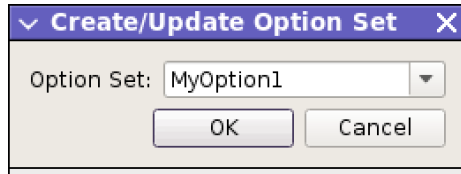


Creating Option Sets

Option sets are retained within the same design session only. If you want to recall the options in a later session, see [Saving Option Sets to a Directory](#).

To create a new option set:

1. In the **Option Sets** area, click . The **Create/Update Option Set** dialog box opens.



2. Type a name in the **Option Set** box.
3. Click **OK** to save the option set to memory and close the dialog box. To save the option set permanently to a file, see [Saving Option Sets to a Directory](#).

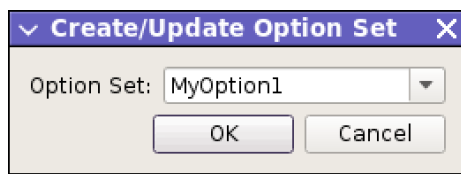
Caution:

If you do not save the option set to a file, you lose your changes when you exit the software.

Modifying Option Sets

To modify an existing option set:

1. In the **Option Set** area, choose the option set that you want to modify.
2. Change the interactive route settings. See [Route Toolbar and Create Route Dialog Box Reference](#).
3. In the **Option Sets** area, click . The **Create/Update Option Set** dialog box opens.

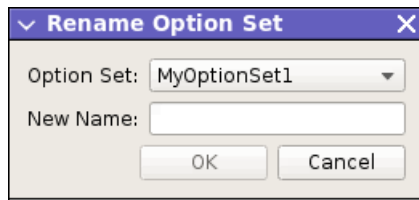


4. Click **OK** to save the changes and close the dialog box.

Renaming Option Sets

To rename an existing option set:

1. In the **Option Set** area, click . The **Rename Option Set** dialog box opens.

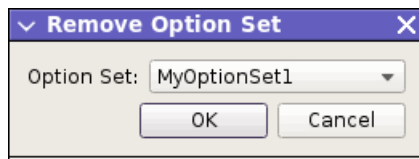


2. Choose the **Option Set** you want to rename.
3. Type a **New Name** for the option set.
4. Click **OK** to save the change and close the dialog box.

Deleting Option Sets

To delete an existing option set:

1. In the **Option Set** area, click . The **Remove Option Set** dialog box opens.



2. Choose the **Option Set** you want to delete.
3. Click **OK** to delete the option set and close the dialog box.

Saving Option Sets to a Directory

To save all existing options to a directory:

1. In the **Option Set** area, click . The **Save Option Sets** dialog box opens.
2. Select the directory you want to use.
3. Click **Choose**. The existing option sets are saved in the directory. Each option set is saved as an XML file.

Reloading Options Sets from a Directory

To load the option sets from a directory:

1. In the **Option Set** area, click . The **Load Option Sets** dialog box opens.
2. Select the directory you want to load.
3. Click **Choose**. The option set names appear in the Option Set list.

Route Toolbar and Create Route Dialog Box Reference

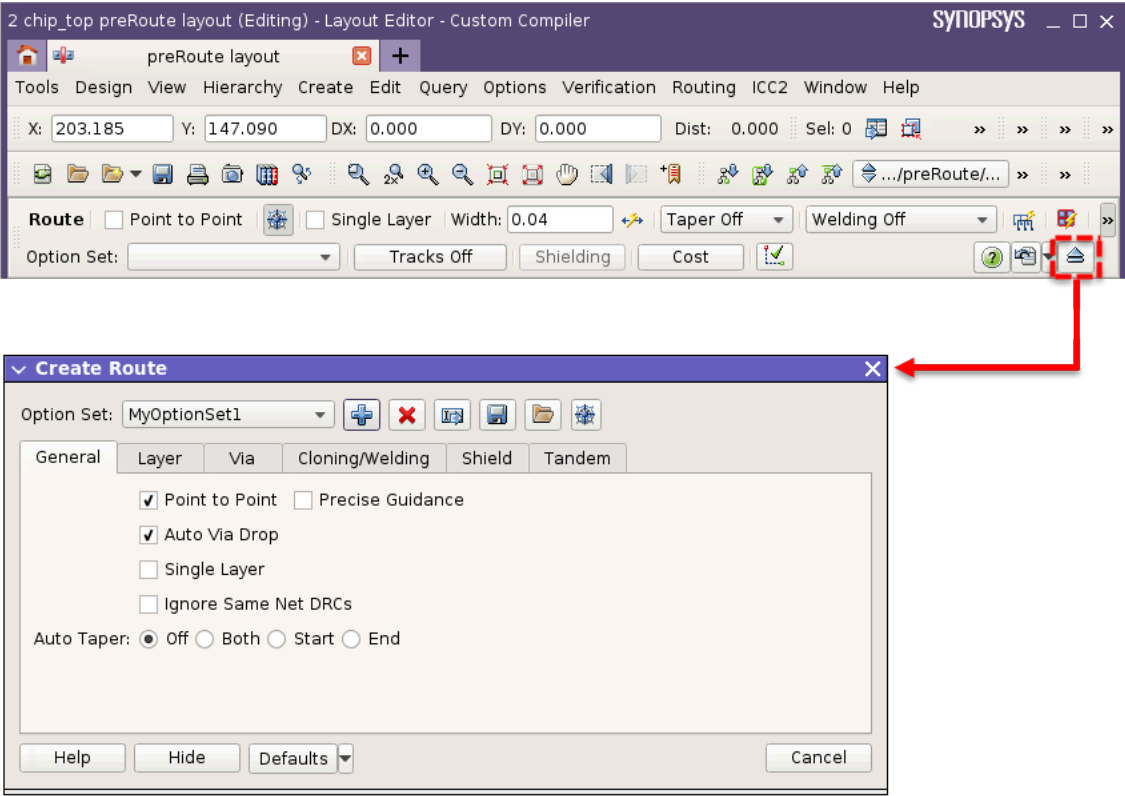
This topic provides a quick reference of the Pre-2020.03 Interactive Router **Route** toolbar and **Create Route** dialog box. For instructions on creating routes interactively, see [Routing Using the Follow-the-Cursor Method](#) and [Routing Using the Point-to-Point Method](#).

The quick reference provides a summary of the following:

- [Basic Options](#)
- [General Tab](#)
- [Layer Tab](#)
- [Via Tab](#)
- [Cloning/Welding Tab](#)
- [Shield Tab](#)
- [Tandem Tab](#)

Basic Options

This section describes the options that appear in both the toolbar and dialog box. In the dialog box, the options appear above and below the tabs.

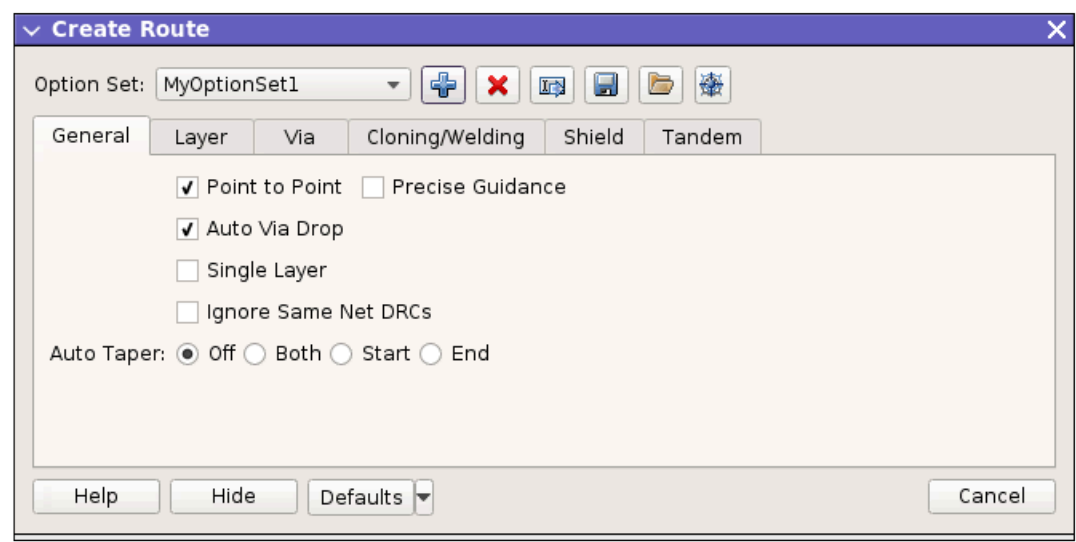


Option	Description
Option Set	Displays the chosen set of options. See Managing Option Sets .
	Displays or hides the Design Navigator so that you can select the nets. See Net-Based Routing Workflow or Color Routing Workflow.
 (toolbar) Help (dialog box)	Displays the Custom Compiler help.
 (toolbar) Defaults (dialog box)	Creates new default options and values, or resets to the selected default options and values. Click the button, and then select one of the following: • Make Defaults for: Uses the current information to create a new set of defaults for cellview, cell, library, or user. • Save: Saves your current preferences as an XML file. • Load from: Loads from the default cellview, cell, library, or user. • Load: Lets you load a previously saved preference file. • Purge from: Lets you delete the default options and values from the cellview, cell, library, or user.

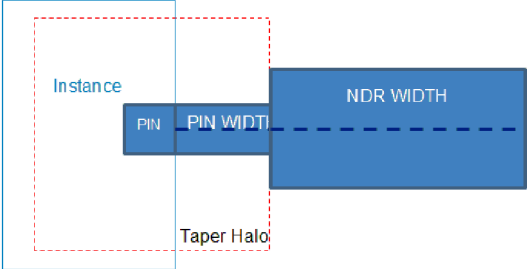
Option	Description
 (toolbar) Hide (dialog box)	Opens the Create Route dialog box. See Route Toolbar and Create Route Dialog Box Reference .

General Tab

The **General** tab and the **Create Route** toolbar contain the following options:



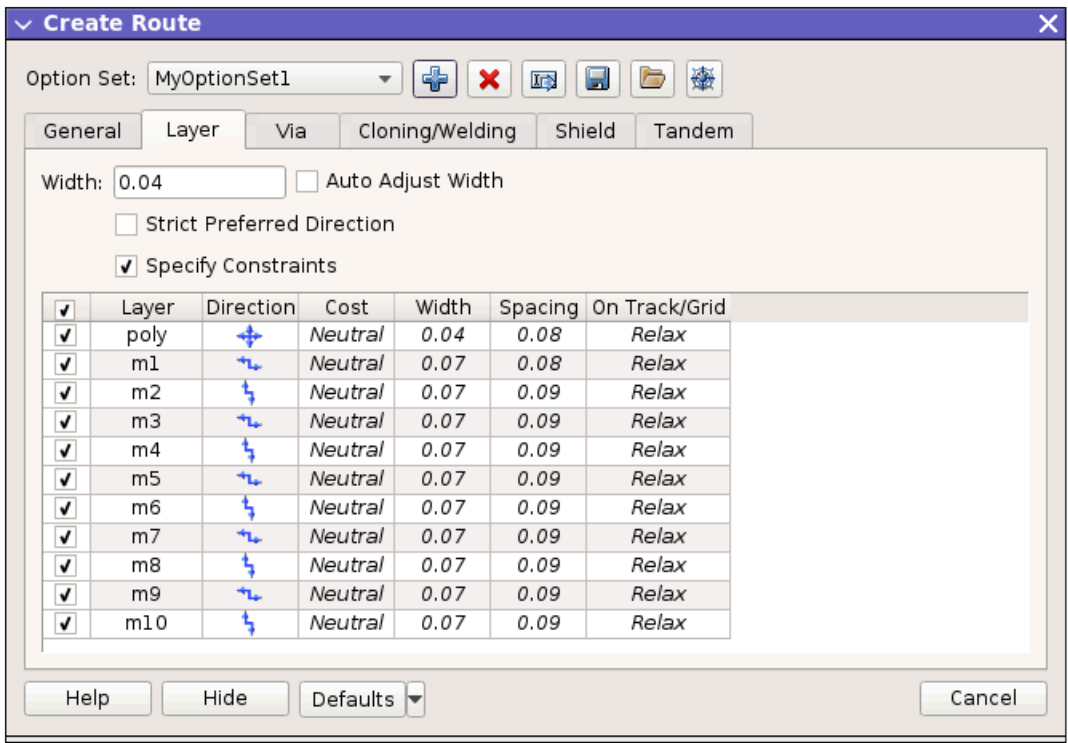
Option	Description
Point to Point	Specifies the interactive routing method: ☐: Creates routes using the follow-the-cursor method. See Routing Using the Follow-the-Cursor Method. ☒: Creates routes using the point-to-point (P2P) method. See Routing Using the Point-to-Point Method. Tcl preference: <code>rtMainModePointToPoint</code>
Precise P2P Guidance (dialog box only)	Provides stricter route guidance than Point to Point routing.
Auto Via Drop	Automatically adds vias to the same net shapes located above or below the current wires being routed. Note the following: This option is typically used with the Auto Welding Mode and Auto Cloning options. For more information, see Adding Vias Automatically. Tcl preference: <code>rtAutoViaDrop</code>
Single Layer	Switches between the multi-layer and single-layer routing modes. - Clear the check box: Specifies the multi-layer mode, in which generated physical interconnect can span multiple layers. - Select Single Layer: Specifies the single-layer mode, in which generated physical interconnect can reside on only one layer.

Option	Description
Ignore Same Net DRCs (dialog box only)	Permits route connections that would otherwise violate same-net DRCs. Select the option when troubleshooting same-net problems. Tcl preference: <code>rtIgnoreSameNetDRCs</code>
Taper mode	Specifies whether the router should perform tapering adjustments (up or down) when connecting to or from a terminal or wire. Auto tapering determines the routing width of the first and last path segment of the routed interconnect based on the dimensions of the level 1 pins and level 0 netted rectangles. • Off: Disables automatic tapering. • Both: Tapers the width of the first and last path segment. • Start: Tapers the first path segment. • End: Tapers the last path segment.  Tcl preference: <code>rtAutoTaperMode</code>

Layer Tab

You can use the **Layer** tab options to specify the valid routing layers and routing directions. By default, the fields are populated with information from the Constraint Editor. Although you can edit the settings and then run an interactive routing operation in the current session, the changes are not written back to the Constraint Editor. Also, if you close the **Create Route** dialog box, your changes are deleted from memory.

The **Layer** tab contains the following options:

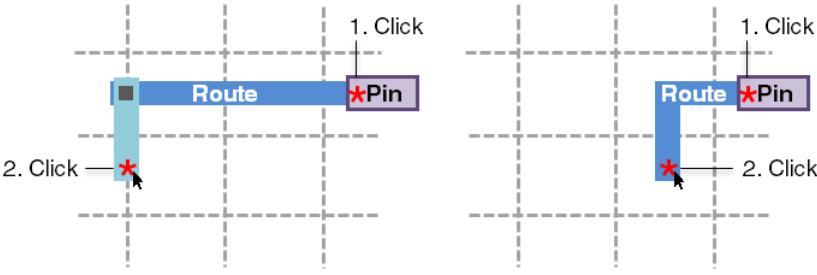


Option	Description
Width	Allows you to specify the width of the route, or displays the current width of the route. The Width box becomes unavailable for the first route segment when you turn on Auto Adjust Width. In this case, you can view the width that is applied. (Auto Adjust Width is off) Type a width value to apply to all layers. (Auto Adjust Width is on) View the width to be applied to the first route segment. After the first route segment, you can override the value and type the desired width in the Width box.] When Auto Adjust Width is on, the width value displayed follows the order of precedence: • The width value previously set in the session. This value will apply to all layers. • An NDR width value associated with the net and relevant layers. Routing width can vary between layers. • A design constraint width value. Routing width can vary between layers. • Foundry rules width. Routing width can vary between layers. For more information, see Adjusting Route Widths Automatically. Tcl preference: leWidth

Option	Description
Auto Adjust Width 	Automatically applies the width that matches the width of the connecting shape and ignores the Width value for the first route segment. After the first route segment is placed, you can change the Width value, or you can press Ctrl+4 to cycle through the following pre-determined widths (values appear in the Width box): - The starting route width, which is based on the pin geometry and route direction. - The greater of the following: the non-default rule (NDR) width value associated with the net, or the current layer's default routing width. For more information, see Adjusting Route Widths Automatically. Tcl preference: <code>leInterconnectAdjustWidth</code>
Strict Preferred Direction	(Multi-Layer only) Creates only preferred-direction routes. Select the option if you want to prevent the router from creating wrong-way routes. For example, in the technology database, the m1 layer definition permits only vertical routes. If you select Strict Preferred Direction, you can only add vertical routes on m1. If you do not select Strict Preferred Direction, you can add vertical and horizontal (wrong-way) routes on m1. Tcl preference: <code>rtStrictPreferredDirection</code>
Specify Constraints	Overrides the design-specific constraints in the Constraint Editor. When you select this option, additional columns appear in the table: Direction, Width, Spacing. The values you enter for these columns override the corresponding values in the Constraint Editor. See Setting Routing Constraints about the Constraint Editor. Tcl preference: <code>rtUseDesignConstraint</code>

Layer table:

Option	Description
Layer	Lists the routing layers. Select the layers that you want to use for routing.
Direction	Specifies the preferred direction: : The router determines the direction. : The preferred direction is vertical. : The preferred direction is horizontal. : The direction will be vertical only. : The direction will be horizontal only. Tcl preference: <code>rtLayerDirection</code>

Option	Description
Cost	Specifies the restrictions to route on the layer: • Low: The cost to route on this layer is less than the cost to route on layers that are neutral. • Neutral: No routing cost bias exists on the layer. • High: The cost to route on this layer is greater than the cost to route on layers that are neutral. The router will use a low-cost layer freely. As the cost increases, the layer usage becomes more restricted.
Width	Specifies the minimum layer width. Tcl preference: <code>rtLayerWidth</code>
Spacing	Specifies the minimum layer spacing. Tcl preference: <code>rtLayerSpacing</code>
On Track/Grid	Specifies whether the route and corresponding parallel shields, jumpers for parallel shields, and bus shields will snap to the wire tracks or the routing grid. Choose one of the following: • On: Snaps routes and shields to active tracks. If the layer does not have active tracks, the routes and shields will snap to the routing grid. If you choose this option, the Cost value is set to 4 automatically, and you cannot change it. When you choose On, routes will not connect to off-track pins. • Off: Does not snap routes and shields to active tracks or grids. Choose this option for gridless routing. • Relax: Snaps routes and shields to active tracks. However, the routes can move off tracks and grids to connect to off-track or off-grid pins. After connecting to off-track pins, the routes will remain off-track until they need to change layer and turn. On Track/Grid = Relax  If tracks and routing grids are not defined and you choose On, the Pre-2020.03 Interactive Router uses an internally determined grid. The grid pitch is route width + minSpacing between shapes. The grid offset is 0 or pitch/2, whichever permits more pins on the grid.

Via Tab

Note:

The via options you set in the Custom Router is universal. The via settings you specify for the Pre-2020.03 Interactive Router is also applied when you use the Auto Router and the Pattern Route Assistant.

The **Via** tab contains the following options:

Option Set: MyOptionSet1

General Layer **Via** Cloning/Welding Shield Tandem

Path Alignment: ☒ Via Center ☐ Via Edge

Preferred Enclosure: ☒ Auto ☐ Cross ☐ Non-cross

Preferred Direction:

Minimum Cuts: 1

☒ Expand

☒	Layer	Via Definition	Cut Class	Direction	Layer1 Enc	L
☒	diff	diffCont	none		0.01 0.03	0.0
☒	diff	pdiff	none		0.01 0.03	0.0
☒	diff	ndiff	none		0.01 0.03	0.0
☒	diff	diffWellCont	none		0.01 0.03	0.0
☒	poly	polyCont	none		0.01 0.03	0.0
☒	m1	via12	none		0.04 0	0.0
☒	m1	via12Line	none		0.04 0	0.0
☒	m2	via23	none		0 0.04	0.0
☒	m2	via23Line	none		0.04 0	0.0
☒	m3	via34	none		0.04 0	0.0
☒	m3	via34Line	none		0.04 0	0.0
☒	m4	via45	none		0 0.04	0.0
☒	m4	via45Line	none		0.04 0	0.0

Help Hide Defaults Cancel

Option	Description
Path Alignment	Aligns the connecting path and via using one of the following methods: • Via Center: Aligns the connecting path's centerline with the center of the via. • Via Edge: Aligns the edge of the connecting path with the edge of the via. Tcl preferences: <code>rtAllowViaOffset</code>
Preferred Enclosure	Specifies how the via's metal layers are positioned: • Auto: The router determines the positions. • Cross: The layers are perpendicular. • Non-cross: The layers are parallel. Tcl preferences: <code>rtViaPreferredEnclosure</code>
Preferred Direction	Specifies the via orientation: • : The router determines the orientation. • : The preferred orientation is vertical. • : The preferred orientation is horizontal. • : The orientation will be vertical only. • : The orientation will be horizontal only. Tcl preferences: <code>rtPrefBarDoubleCutViaOrient</code>, <code>rtStrictFollowPrefBarDoubleCutViaOrient</code>
Minimum Cuts	The minimum number of cuts in the via. Tcl preferences: <code>rtMinimumCut</code>

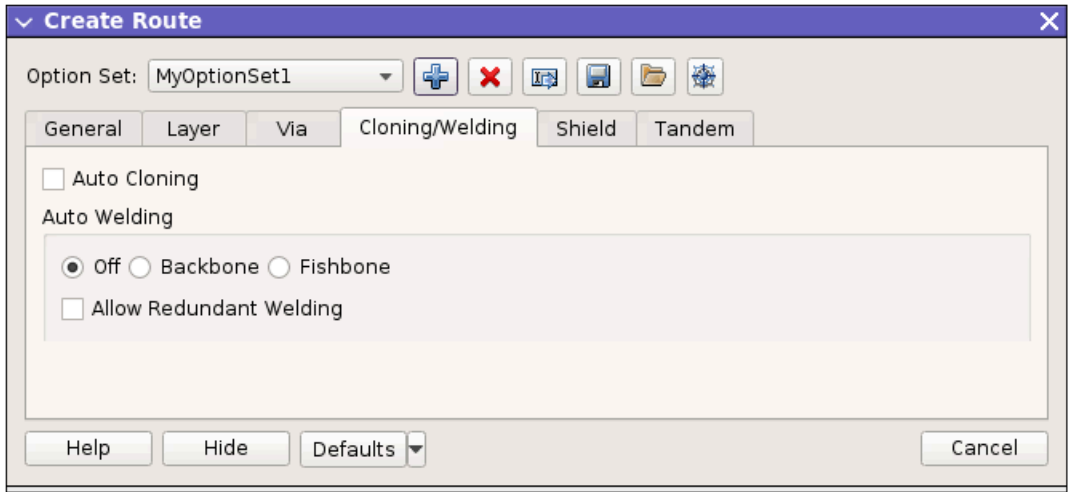
Layer table:

Option	Description
Expand	Displays or hides additional columns in the layer table: Direction, Layer1 Enc, and Layer2 Enc. Tcl preferences: <code>rtUseExpandedViaSetting</code>
Layer	Lists the layers.
Via Definition	Specifies the via definition to use when adding the via on the specific layer.
Cut Class	Specifies the cut class to use when adding the via.

Option	Description
Direction	Specifies the via direction (or orientation). : The router determines the orientation. : The preferred orientation is vertical. : The preferred orientation is horizontal. : The orientation will be vertical only. : The orientation will be horizontal only.
Layer1/2 Enc	Amount to extend layer1 and layer2 beyond the cut edges in the X and Y direction.

Cloning/Welding Tab

The **Cloning/Welding** tab contains the following options:



Option	Description
 Auto Cloning	Automatically replicates routes and connections between rows of devices. This option is typically used with the Auto Via Drop and Auto Welding Mode options. For more information, see Cloning Routes Automatically . Tcl preference: <code>rtAutoCloning</code>

Option	Description
Welding mode	Automatically branches (taps off) from the main interconnect (trunk) to connect to nearby pins on the same net in the Single Layer or Multi Layer modes. The branches can be above or below the main path. This option is typically used with the Auto Via Drop and Auto Cloning options. See Connecting to Neighboring Devices Automatically. Tcl preference: <code>rtAutoWeldingMode</code>
Allow Redundant Welding	Automatically connects to a set of adjacent pins that the router has determined to be already connected. Tcl preference: <code>rtAllowedRedundantWelding</code>

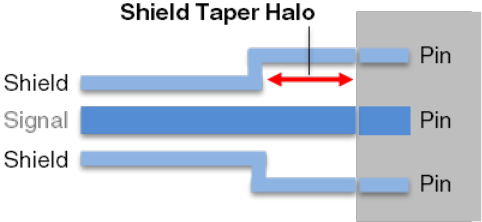
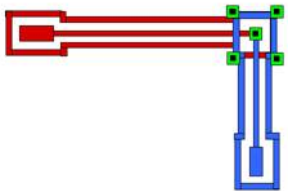
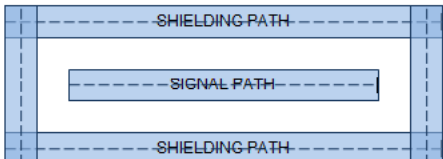
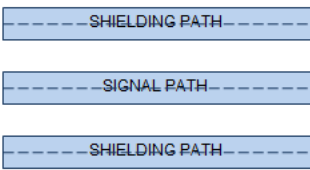
Shield Tab

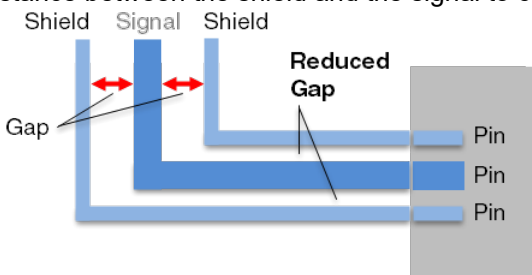
For instructions on how to add shielding, see [Creating Routing Constraints](#).

The **Shield** tab contains the following options:

The screenshot shows the 'Create Route' dialog box with the 'Shield' tab selected. The 'Option Set' is 'MyOptionSet1'. The 'Shield Nets' field is empty. 'Taper Halo' is 0. 'Enclose' has checkboxes for 'Pin' and 'Via', both checked. 'Allow Wrong Way' is checked, 'Connect to Aligned Pin' is unchecked. 'Sharing' is checked, 'Use Via Array for Coaxial' is unchecked. 'Jumper Spacing (um)' has fields for 'Tandem: 0', 'Parallel: 0', and 'Coaxial: 0'. 'Minimum Length (um)' has fields for 'Signal Segment: 0' and 'Connected Shield: 0'. 'Connect to Supply' is unchecked. 'Min Shield Length' is 0 um. 'Connect at Mesh Overlap' is unchecked. 'Layer Offset' has radio buttons for 'Above', 'Either' (selected), and 'Below'. 'Connect with Route' is unchecked. 'Tie Spacing' is 0 um and 'Search Distance' is 0 um. 'Report Coverage' is unchecked. At the bottom are 'Help', 'Hide', 'Defaults' (dropdown), and 'Cancel' buttons.

Option	Description
Shield Nets	Specifies the net to use as shield. The router will use this value only if the information is not provided in the Net Shielding Constraint. Tcl preferences: <code>rtShieldNet</code> , <code>rtShieldSecondNet</code>

Option	Description
Taper Halo	Specifies the distance between the pin and the location at which the shield must taper. Default: 0.0 (tapers the shield between the pin and  the first turn.) Tcl preference: <code>rtShieldTaperHalo</code>
Enclose: Pin	When selected, extends the parallel shield line to wrap around a pin if  there is enough room. Tcl preference: <code>rtShieldEnclosePin</code>
Enclose: Via	When selected, the parallel shield line is extended to wrap around a via. Tcl preference: <code>rtShieldEncloseVia</code>
Allow Wrong Way	Adds shielding paths in one of the following ways: - Select the option to add shielding paths in either of the two directions across the routing layers.  - Clear the check box so that all shielding paths are oriented in the same direction across all routing layers.  If a layer permits bi-directional routing, the router can create shielding paths in both directions, but the segments will have open ends. Tcl preference: <code>rtAllowWrongWayShield</code>

Option	Description
Connect to Aligned Pin	Reduces the distance between the shield and the signal to connect to  the shield pins. Tcl preference: <code>rtConnectShieldToAlignedPin</code>
Sharing	Permits shield to be shared among signal nets in the same group that have parallel and split coaxial shield styles. Shield Sharing does not apply to other shield styles. For more information, see Shared Shielding. The Shield Sharing option is also available in the Add Shields dialog box and the P2P Shielding Options dialog box. Enabling or disabling Shield Sharing in one of the dialog boxes automatically enables or disables the option in other dialog box. For more information, see Setting Options in the Add Shield Dialog Box and Setting Options in the P2P Shield Dialog Box. To apply shield sharing to a specific group of nets, use the Sharing setting in the Net Shielding constraint. For information about the Constraint Editor, see Setting Routing Constraints. To route wires close to power straps and use the power straps as shared shielding, set the <code>rtUsePowerStrapAsShield</code> preference to <code>true</code>. This preference is applicable for auto routing only.
Use Via Array for Coaxial	Uses the via array to tie solid coaxial shields instead of jumper vias. Tcl preference: <code>rtShieldUseViaArrayForCoaxial</code>
Jumper Spacing	Specifies the spacing between consecutive jumper pairs that connect different types of shields: Tandem, Parallel, and Coaxial. If the value is not set or is 0, no jumper or jumper via is added. For the descriptions of the different shield types, see Shielding Styles and Behavior. Tcl preferences: <code>rtShieldTandemJumper</code>, <code>rtShieldParallelJumper</code>, <code>rtShieldCoaxialJumper</code>

Option	Description
Minimum Length (um)	Specifies the minimum length of signal segments and associated shield segments. If a signal or connected shield segment is shorter than the specified value, the router will not let you create a shield segment for it. • Signal Segment: The minimum length of the signal segment. The default value is 0 um. The Minimum Segment control in the shield constraint can also specify the minimum signal shield length. Note that point-to-point (P2P) shielding can shield part of a signal segment. Therefore, minimum signal segment has no impact on P2P shielding. • Connected Shield: The minimum length of connected shield segments. The default value is 0 um. Tcl preferences: <code>rtShieldMinSignalSeg</code>, <code>rtMinConnectedShieldLength</code>
Connect to Supply	Lets you specify requirements for how shields will connect to the nearest power mesh. Tcl preference: <code>rtConnectShield</code> • Min Shield Length (um): Specifies the minimum shield length required to connect to the power supply. Tcl preference: <code>rtMinShieldSupply</code> • Connect at Mesh Overlap: Creates multiple shield connections to the power mesh for a shield. Creates a shield via at each intersection of a shield path segment and a power strap. If a shield does not intersect with any power strap, the default four-pass connecting process is followed. Tcl preference: <code>rtShieldConnectAtMeshOverlap</code> • Layer Offset: Specifies the power mesh layer to connect to the shield: Above, Either above or below, or Below the shield layer. The options are available only if you enable Connect At Mesh Overlap. Tcl preference: <code>rtShieldConnectLayerOffset</code> • Connect with Route: Connects shields to ground. This option creates long ties that might block future routing. Select this option only if you must connect shields to ground using the Pre-2020.03 Interactive Router. Note that these shields are marked as dynamic shields in ICC designs. Tcl preference: <code>rtShieldConnectWithRoute</code> • Tie Spacing (um): The distance, in micrometers, between wire-tie connections along a shield path segment. Tcl preference: <code>rtShieldTieSpacing</code> • Search Distance (um): The distance that the Pre-2020.03 Interactive Router uses when searching for targets to tie. Tcl preference: <code>rtShieldTiePowerSearch</code>
Report Coverage	Creates a shield coverage report at the end of the auto-route run. The coverage can be larger than 100% if you selected Enclose Pin or Enclose Via, and any shielding added in the back of the via is considered extra shielding.

If you want to connect shielding to VSS instance pins, you can set the `rtConnectShieldToAlignedPins` preference to `true`.

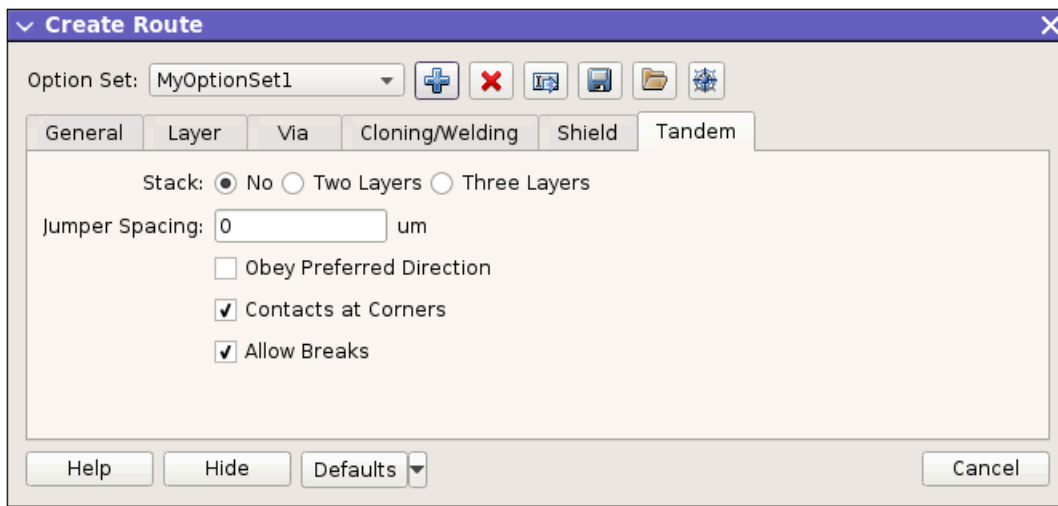
Shield vias are automatically added when you add shielding. By default, the router will use the corresponding signal's via definitions for shield via creation. If, for some reason, you want to use different via definitions when adding shield vias, you can do one of the following:

- Choose the via definitions to use in the Net Shielding constraint. For information, see [Net Shielding Constraint](#).
- Set the `rtSyncShieldViasWithSignalVias` Tcl preference to `false`. For information about the Tcl preference, see the *Custom Compiler Tcl Reference Manual*.

For more information about shield vias, see [Shield Vias](#).

Tandem Tab

The **Tandem** tab contains the following options:



Option	Description
Stack	Specifies the number of layers for the tandem route to create: • No: Stacking is not permitted. • Two Layers: if no tandem constraint is given, the redundant tandem route will be placed above the route to be realized. • Three Layers: if no tandem constraint is given, the redundant tandem routes will be placed above and below the route to be realized. If the route is started on the lowest layer, the three layer tandem route will not be realized at the start; once the route distances itself from the start point, the three layer tandem route can be successfully used.
Jumper Spacing	The distance between jumper vias used to connect tandem routing metal segments.
Obey Preferred Direction	Enables partial tandem connections while ignoring preferred direction settings.
Contacts at Corners	Inserts vias at route corners (such as on layer transitions).
Allow Breaks	Allows partial tandem connections. If disabled, only full tandem connections can be realized.

2

Interactive Routing Cursor Movement and Behavior

Routing results depend on how you move the pointer on the layout canvas and how the Pre-2020.03 Interactive Router interprets the movements.

This section describes interactive routing cursor movements in the following cases:

- [Basic Interactive Cursor Movements](#)
- [Interactions with Obstructions](#)
- [Routing Over and Beyond Objects](#)
- [Routing Around Obstacles in Trackless Designs](#)
- [Adding Points Along the Route](#)
- [Adding Vias](#)
- [Unroutable Objects](#)
- [Jogs and Via Costs in Routing](#)
- [Self-Intersecting Routing Patterns and Same-Net Obstructions](#)

Basic Interactive Cursor Movements

When you begin to move the cursor on the canvas after choosing the **Routing > Interactive Route** command, the router displays a DRC-correct route from the first point you add on the canvas to the current cursor position. As you continue to move the cursor, the router computes and displays the best possible legal route, based on layer constraints, preferences, and options.

Figure 1 shows an example of tie breaking for routes of equal cost.

Figure 1 Tie Breaking Route

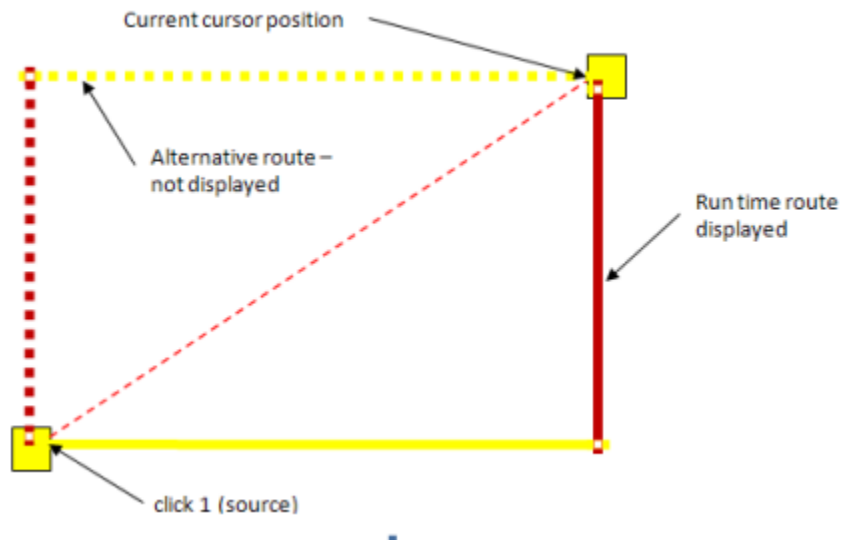
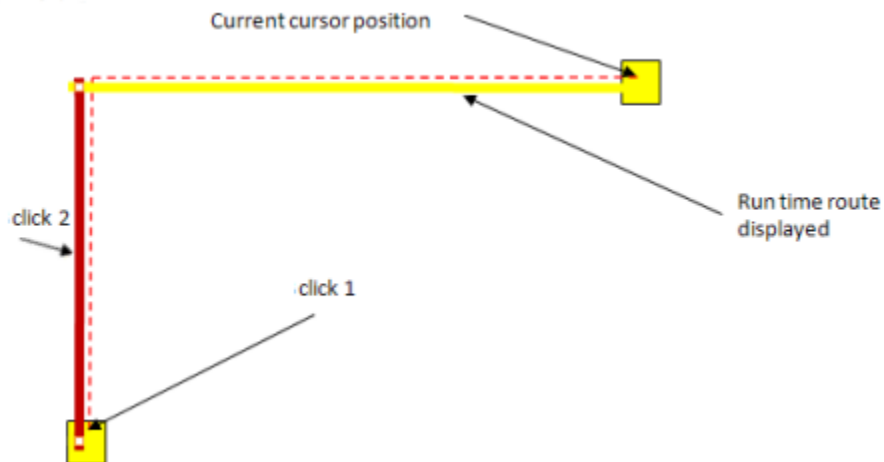


Figure 2 shows how you could insert an intermediate point for the situation displayed in Figure 1, if you wanted the alternative route.

Figure 2 Alternate Routes



Interactions with Obstructions

Figure 3 through Figure 7 demonstrate examples of interactions between design obstructions and generated routes.

Figure 3 *Interaction with Blockage*

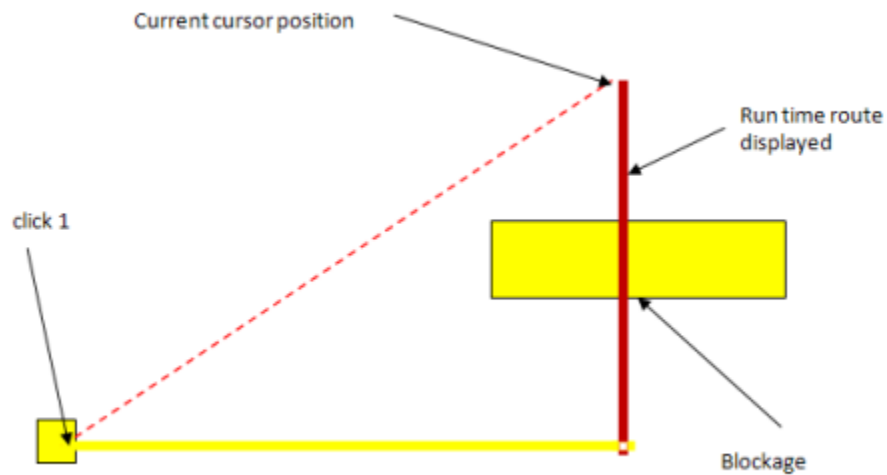


Figure 4 *Interaction with Blockage*

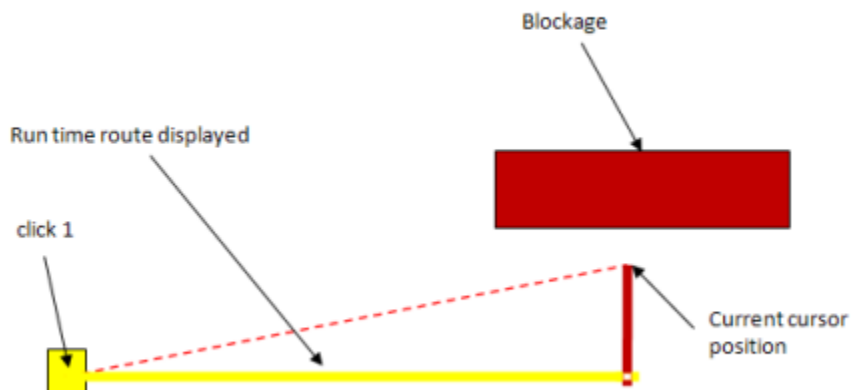


Figure 5 shows how the path followed by the mouse does not affect the displayed run-time pool route. However, as the cursor moves from the situation in Figure 4 to the situation in Figure 5, the displayed route will jump from the route displayed in Figure 4 to the route displayed in Figure 5.

Figure 5 *Interaction with Blockage*

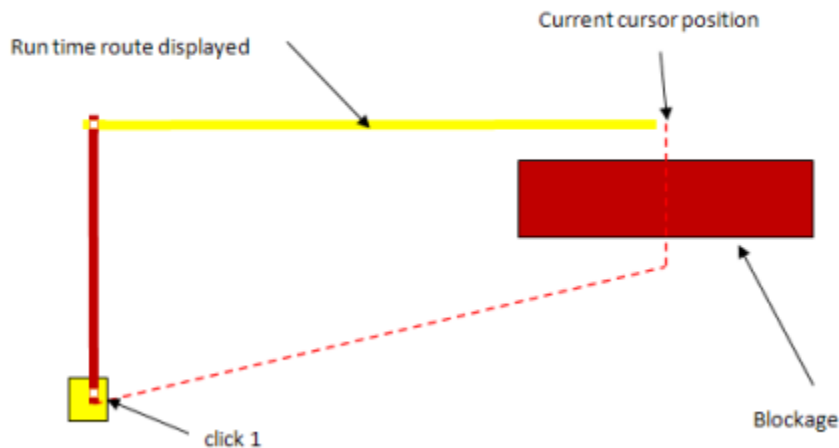


Figure 6 *Interaction with Blockage*

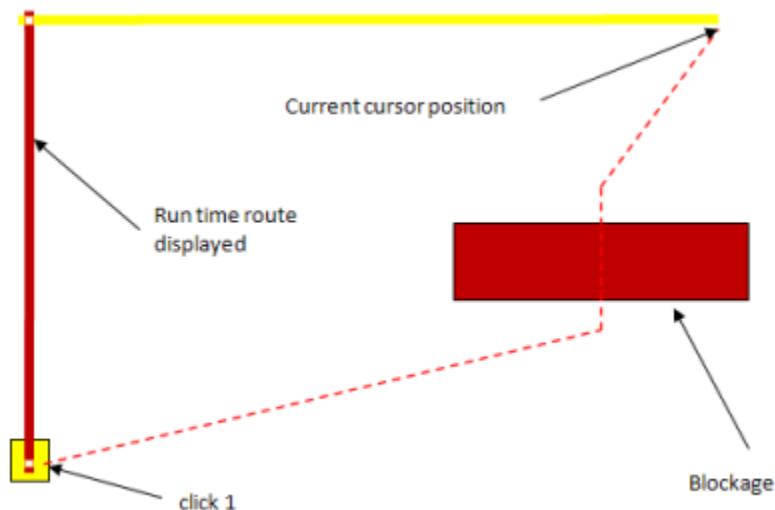
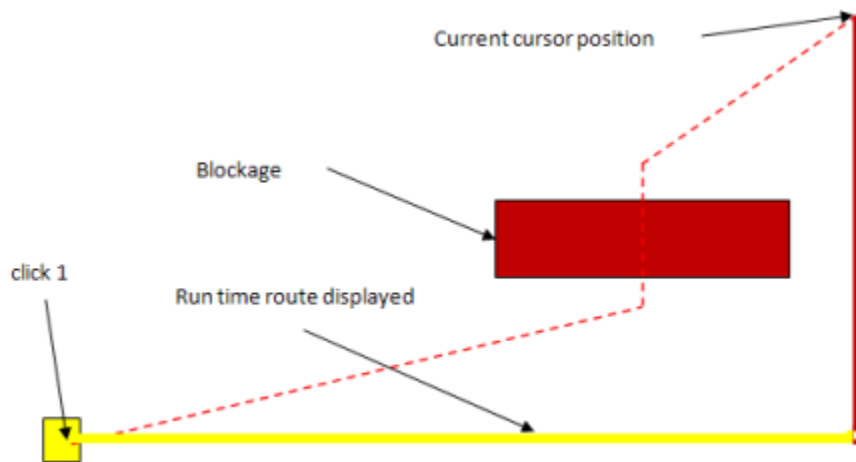


Figure 7 Interaction with Blockage



Routing Over and Beyond Objects

When routing over an object that is compatible for connectivity (for example, connectivity aware mode is off or object is not netted), the connection will be made to the active object under consideration ([Figure 8](#)).

When the cursor moves off the object's geometry, the objects will be considered to be an obstruction to be avoided ([Figure 9](#)).

Figure 8 Over Compatible Geometry

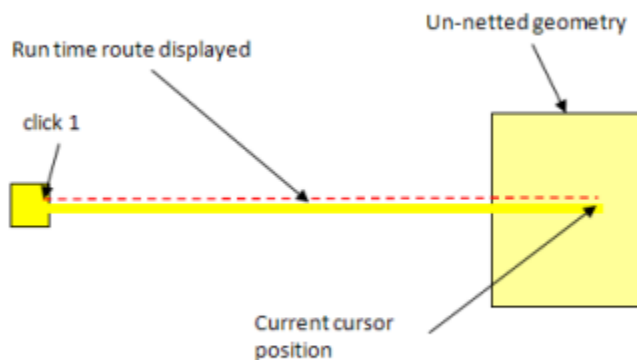
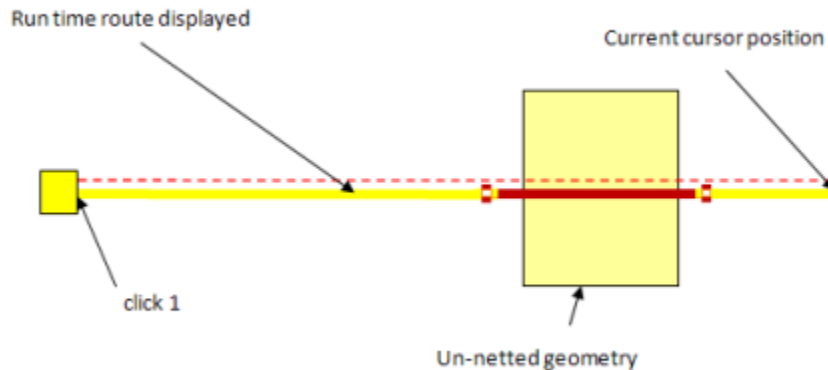


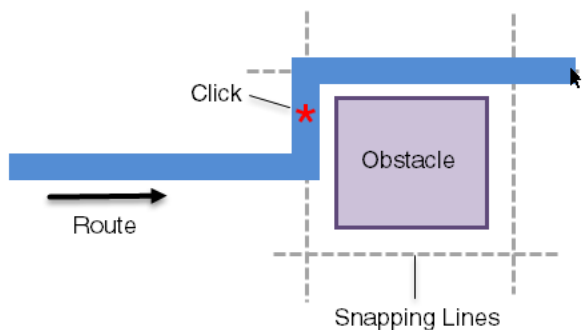
Figure 9 Over Compatible Geometry and Beyond



Routing Around Obstacles in Trackless Designs

In trackless designs, the Custom Router provides invisible snapping lines so that routes can be placed close to obstacles without violating spacing and other design rules. By eliminating time-consuming guesswork near the obstacles, the snapping lines can facilitate and accelerate routing tasks.

During interactive routing, as a route approaches an obstacle, it is snapped to the nearest snapping line surrounding the obstacle. As the route moves around the obstacle, a click along the first side ensures that the router retains its computation. As the route continues, the snapping lines ensure that the route maintains a legal distance.

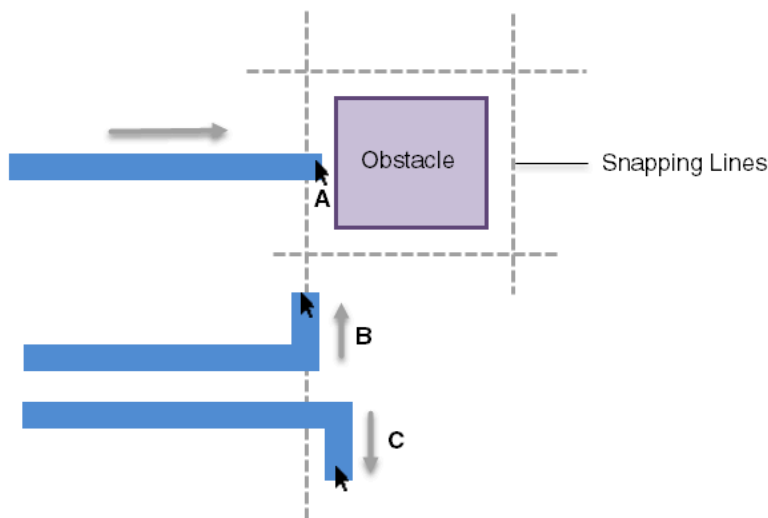


Routes are snapped to snapping lines when:

- The route pointer is between the snapping line and the edge of an obstruction (**A**).
- The route pointer is not between the snapping line and the edge of an obstruction, but it is moving toward the obstruction (**B**).

Snapping does not occur when both of the following conditions are met (**C**):

- The route pointer is not between the snapping line and the edge of the obstruction.
- The route is moving away from the obstruction.



Adding Points Along the Route

An *intermediate add-point event* in follow-the-cursor interactive routing mode is an add-point event that does not start or end the current interactive routing operation.

An add point event can be triggered by one of the following:

- A single click on the canvas
- Hitting the space bar (standard binding)
- A right-click on the canvas (click-and-menu interaction)

For follow-the-cursor interactive routing, an intermediate add point event results in the *freezing* of the route computed so far. The route however will not be committed to the database at this point; it will still reside within the run-time pool memory. The *frozen* route is displayed at the time of the add point event. You can undo this action by removing the last added point (using the Remove Point command or press Backspace). After the add-

point event, the editing mode will continue to be interactive follow-the-cursor routing. The new routing source for the continuation of the routing operation will be the last segment of the route *frozen* so far. You can start the next leg of the routing operation from any point along the last *frozen* segment (Figure 10).

Figure 10 Add Point Event

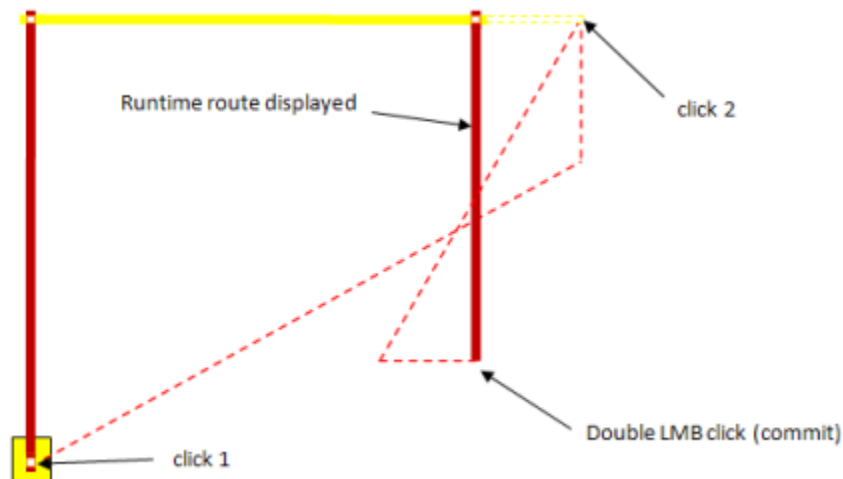
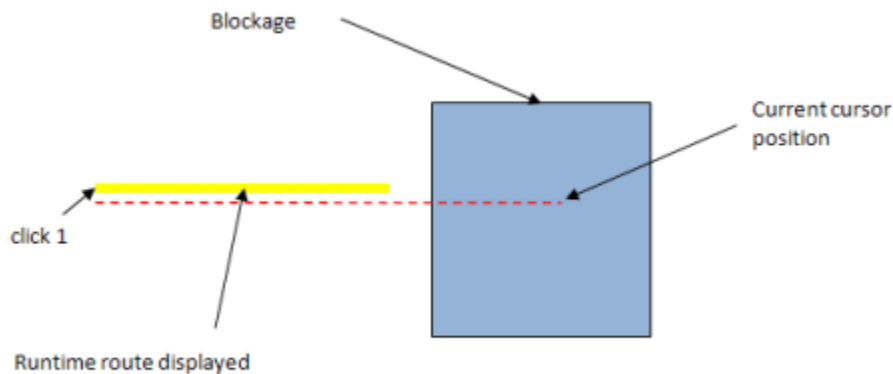
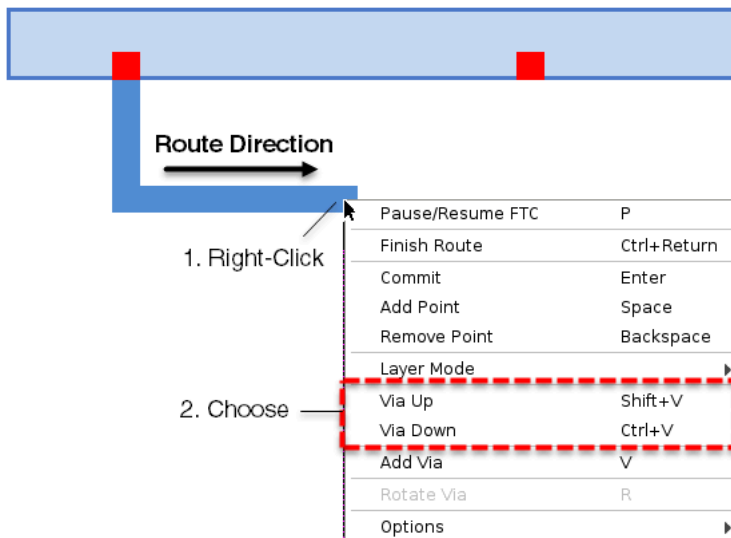


Figure 11 No Legal Solution

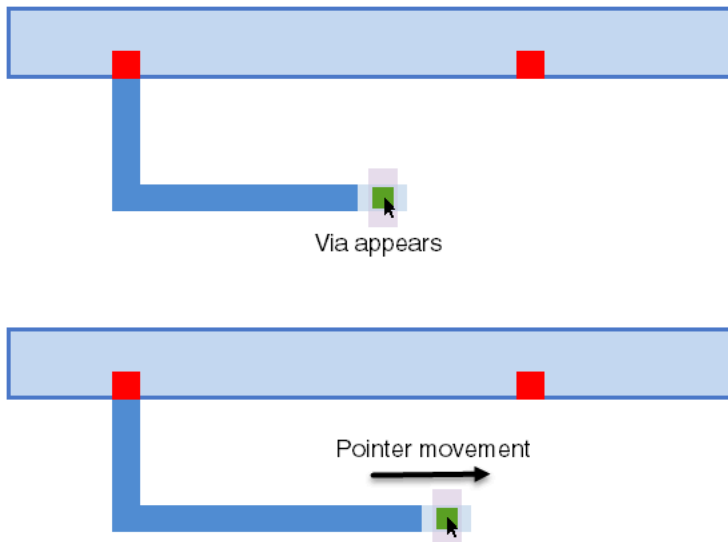


Adding Vias

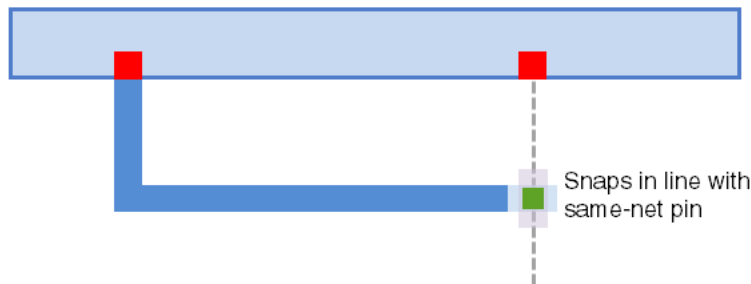
During follow-the-cursor single-layer interactive routing, you can right-click and choose **Via Up** or **Via Down** to add a via above or below the current layer. See [Adding Vias and Making Route Adjustments](#).



As soon as you choose **Via Up** or **Via Down**, a standard via appears under the pointer, at the end of the current route. The via moves with the pointer.



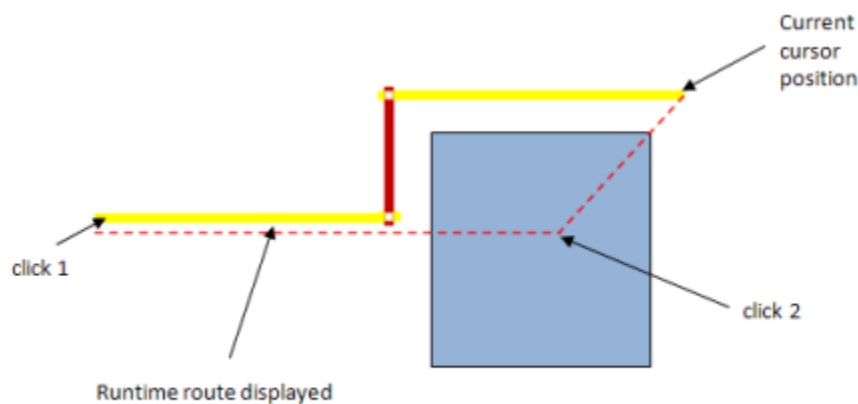
As you move the via close to a same-net pin that is on the same layer, the via snaps to a location directly in line with the pin. The snapping eliminates any position guesswork and facilitates the routing and via placement process.



To add the via, click the canvas, or right-click and choose **Add Point**. See [Adding Vias and Making Route Adjustments](#).

Unroutable Objects

The router does not add routes to unroutable objects. For example, in the follow-the-cursor mode, If you click to enter a point at an unroutable object, the router will add a route around the unroutable object to bypass it.



Jogs and Via Costs in Routing

[Figure 12](#) demonstrates the intended behavior when the router is in the follow-the-cursor mode with default preference costing parameter values. If the cursor is moved in such a way as to require a one track jump in the route, a wrong way jog on the same layer will be

used rather than switching to the adjacent layer. However if the jump/jog is more than one track, as shown in [Figure 13](#), the router will route that segment using an adjacent layer rather than a two+ track long wrong-way jog. If you would prefer to see no routing layer change in the routing pattern, you can specify the behavior by selecting either the **Single-Layer** or **Multi-Layer** option from the **Route** toolbar or menu ([Figure 14](#)).

Figure 12 One-track Routing Jog

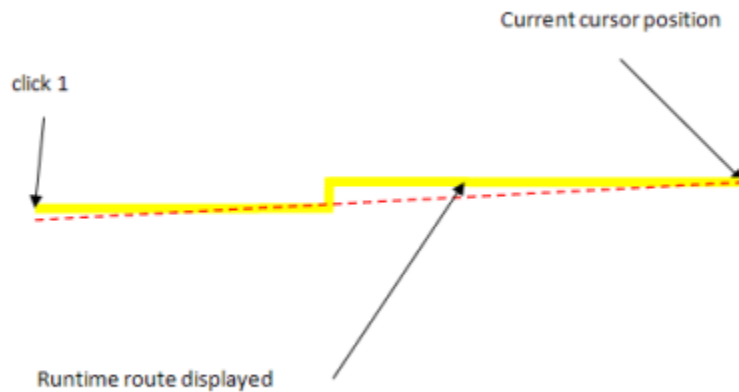


Figure 13 Multi-track/Multi-Layer Routing Jog

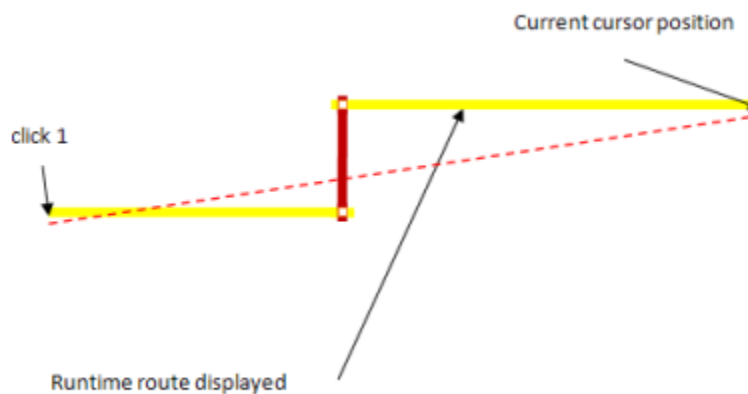
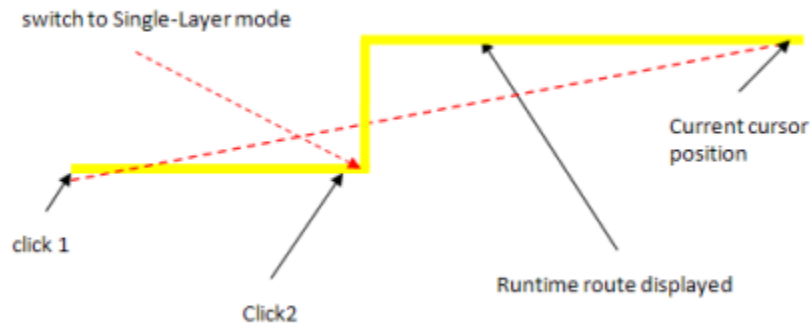
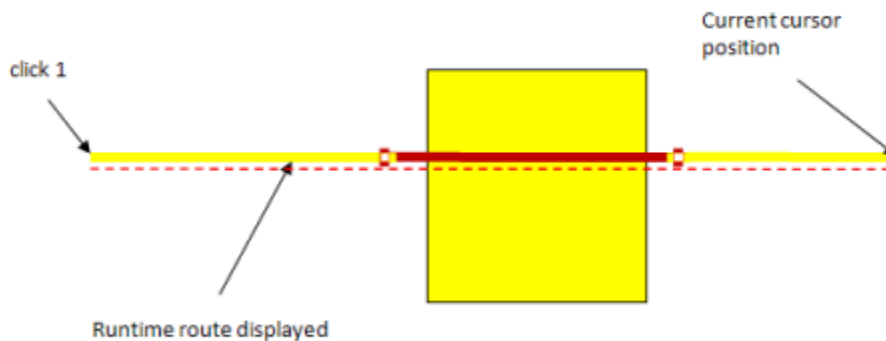


Figure 14 Single-Layer Multi-track Routing Jog



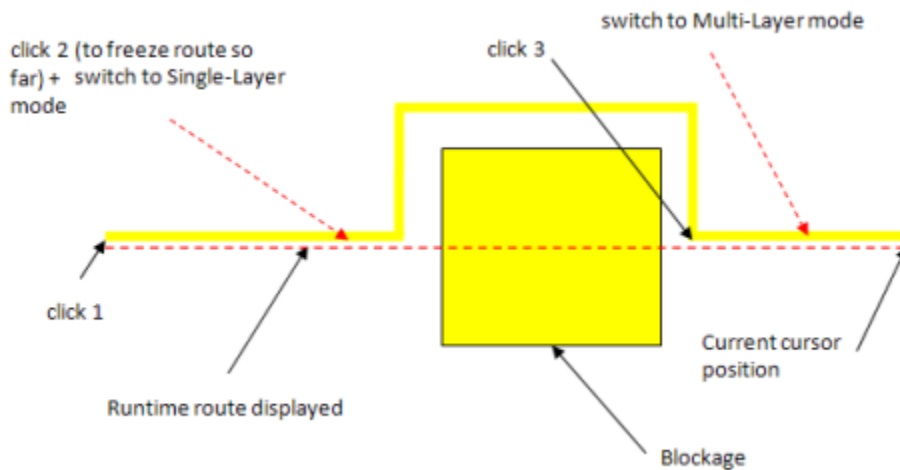
Having minimum parameter values for via and jog costs set by default means that the routing engine, in the follow-the-cursor mode, will almost always perform a via up or down operation to avoid an obstruction, if permissible. [Figure 15](#) illustrates the default behavior in the follow-the-cursor multi-layer routing mode for the mouse movement. The router will automatically create the wrong way wire on M3 in order to clear the M2 obstruction.

Figure 15 Wrong Way Routes to Avoid Obstruction



In order to realize the routing illustrated in [Figure 16](#), you have to use the menu to switch to the Single-layer mode while the route is being created, and then switch back to Multi-layer mode.

Figure 16 Same layer Jogs



Note:

In [Figure 16](#), click 3 is necessary to force the routing to hug the blockage geometry in this specific example.

If the routing automatically generated in the example in [Figure 17](#) is not the desired pattern, you can generate the pattern in [Figure 18](#) through the actions shown. Similarly, the pattern illustrated in [Figure 18](#) can also be realized easily through cursor movements and clicks.

Figure 17 Same Layer Jogs

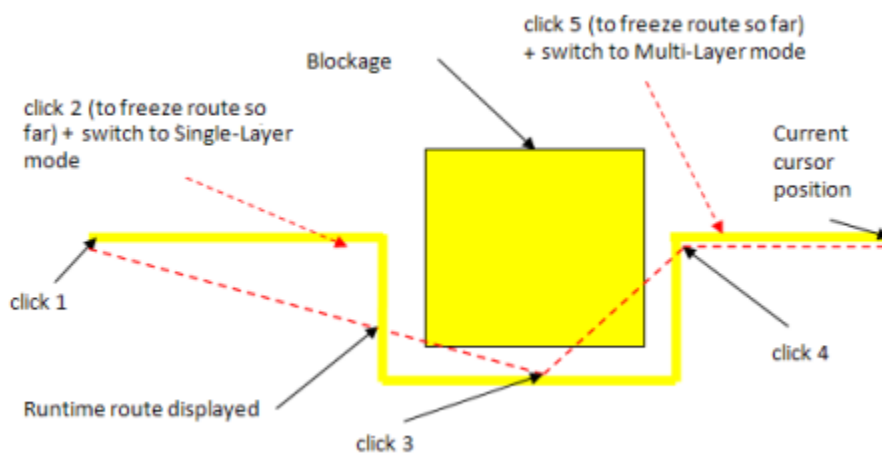
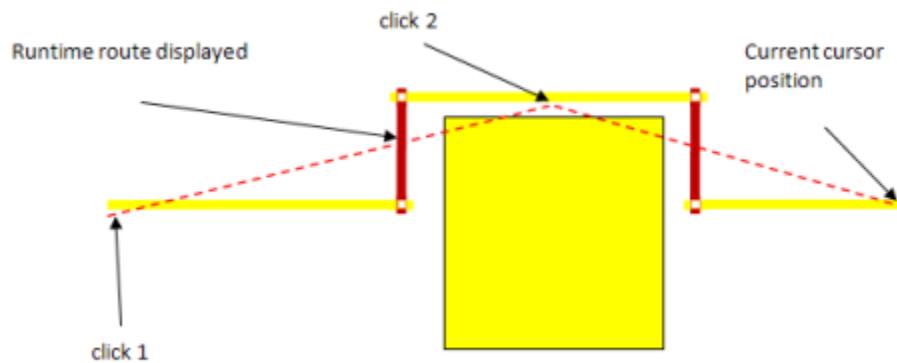


Figure 18 Wrong-way Jogs Minimized



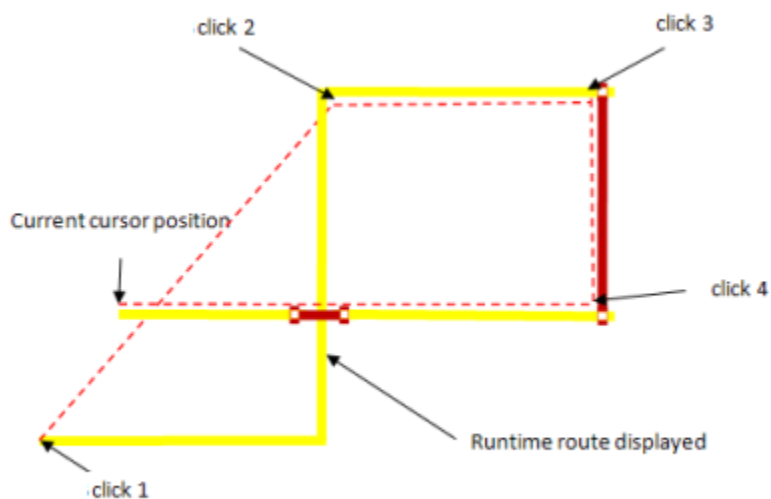
Self-Intersecting Routing Patterns and Same-Net Obstructions

The router treats same net wiring in the runtime pool (prior to a commit) as obstructions to be avoided.

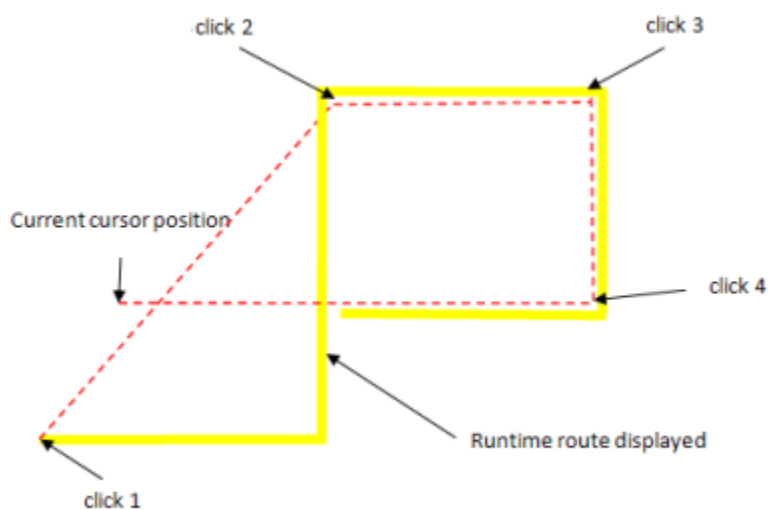
Note:

It is possible to tap into same net wiring when the wiring is not in the run time tool; that is, same net wires, vias, and pins were not created during the current routing operation. The following example shows same-net wiring being treated as a routing obstruction in follow-the-cursor multi-layer mode and how, in this case, the pattern would be routed by the routing engine

Chapter 2: Interactive Routing Cursor Movement and Behavior Self-Intersecting Routing Patterns and Same-Net Obstructions



The following example shows how treating same-net wiring in the runtime pool as obstructions results in a truncated wiring pattern for the example depicted (follow-the-cursor single-layer mode).



3

Troubleshooting Error Messages

This appendix presents the list of Pre-2020.03 Interactive Router error messages, potential causes of the errors, and ways to resolve the errors.

This appendix contains the following topics:

- [Where the Error Messages Are Displayed](#)
- [Types of Error Messages](#)
- [IGCR_CREATE_ROUTE Error Messages](#)

Where the Error Messages Are Displayed

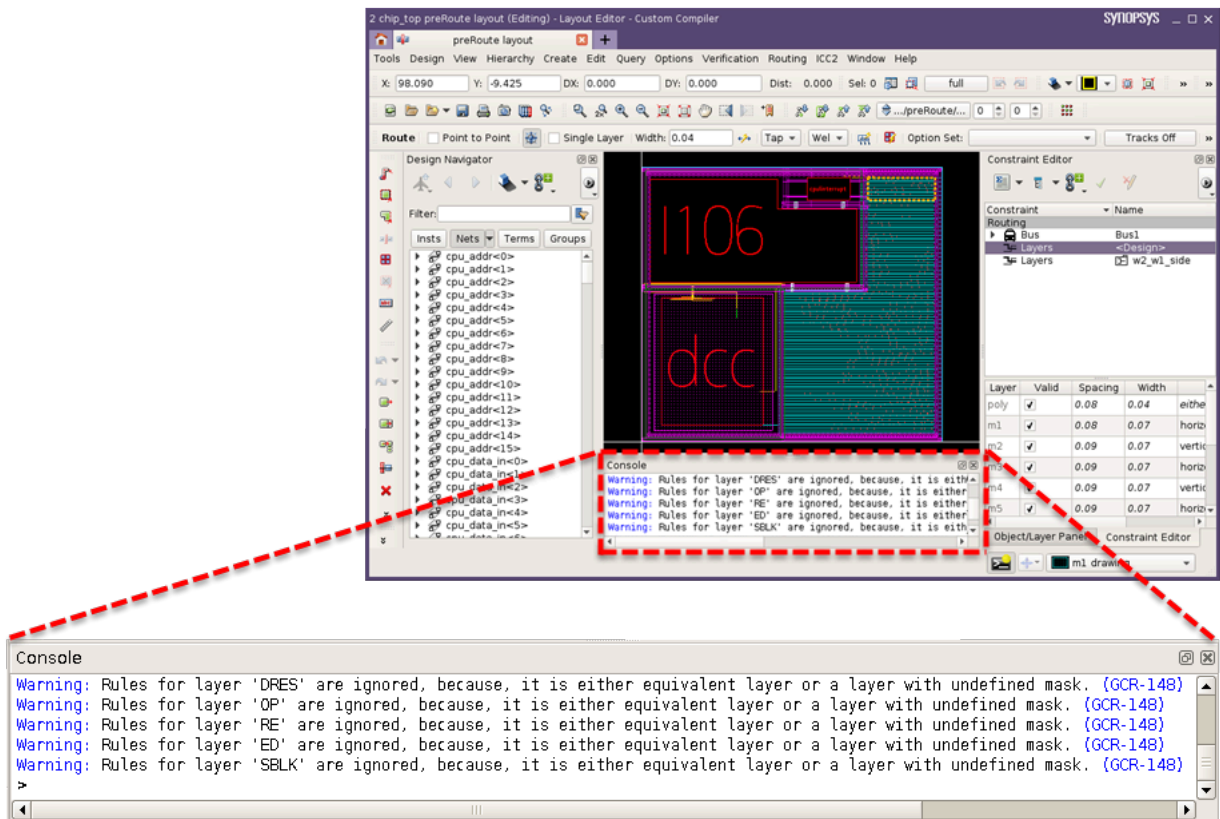
The Custom Router displays error messages in the following locations:

Console

Messages can appear in the Console when you:

- Enable either the Auto Router or the Interactive Router
- Start Auto Routing or Interactive Routing, and while routing is in operation

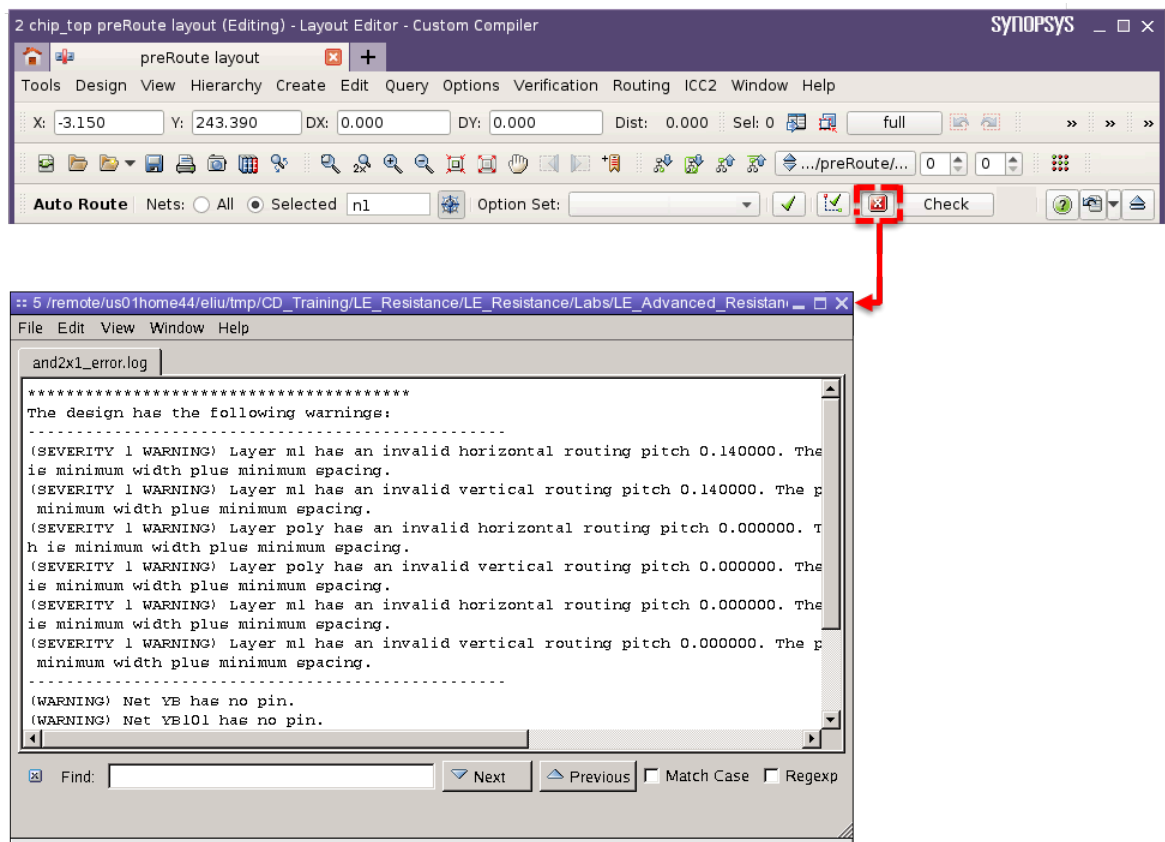
For information about the Console tab and the Console pane, see the *Custom Compiler Environment User Guide*.



For the descriptions of the message types, see [Types of Error Messages](#). For the list of error messages, their causes, and potential solutions, see [Troubleshooting Error Messages](#).

Text Viewer

Messages appear in the Text Viewer when you run the Data Checker in preparation for auto routing. For information about the Text Viewer, see the *Custom Compiler Environment User Guide*.



For the descriptions of the message types, see [Types of Error Messages](#). For the list of error messages, their causes, and potential solutions, see [Troubleshooting Error Messages](#).

Types of Error Messages

The types of messages are as follows:

Type of Message	Description
Error	One or more problems have occurred and the operation is unable to continue. You must correct the problems so that the operation can continue or finish successfully. In some cases, you might have to restart the operation.
Warning	One or more conditions might produce unexpected results or cause problems in the future. Although the operation was able to continue, you might want to follow the recommended corrective actions to change the results or prevent problems in the future.

Type of Message	Description
Information	Informational messages provide notifications only. Corrective actions are typically not necessary. For example, an informational message might let you know that an operation is in progress or has finished.

For the list of error messages, their causes, and potential solutions, see [Troubleshooting Error Messages](#).

IGCR_CREATE_ROUTE Error Messages

The following table lists:

- The error and warning messages you might encounter when using the Pre-2020.03 Interactive Router. The messages are listed alphanumerically by message ID.
- The type of message: error or warning. See [Types of Error Messages](#).
- The causes of the errors or warnings.
- Ways to resolve the errors or warnings.

If you are still experiencing problems after trying the solutions, or if an error not on the list is displayed, contact [Synopsys Support](#).

Note:

Because Information messages are self-explanatory and do not require corrective actions, they are not listed in the following table.

ID	Type	Message	Causes	Solutions
IGCR_CREATE_ROUTE-002	Warning	Could not commit in this state.	The Interactive Router has encountered an invalid state and cannot commit the routing.	Contact Synopsys Support.
IGCR_CREATE_ROUTE-003	Warning	Could not set new width because %d is invalid width.	Width may be an odd number, or in violation of the <code>minWidth</code> or <code>maxWidth</code> constraint.	Check that the width is an even number and is within the range of the <code>minWidth</code> and <code>maxWidth</code> constraints.
IGCR_CREATE_ROUTE-009	Warning	There is no valid and visible Routing LPP on the OLP.	There is no valid and visible routing layer-purpose pair in the Object/Layer Panel.	Check that the appropriate layer-purpose pair is visible and selected for routing in the Object/Layer Panel Assistant.

ID	Type	Message	Causes	Solutions
IGCR_CRE ATE_ROUTE E-020	Warning	ISBR does not support shielding constraint on net %s. Constraint will be ignored.	The shielding constraint is ignored.	Check the Net Shielding Constraints for the specified nets.
IGCR_CRE ATE_ROUTE E-023	Warning	Width %.10g is not an even multiple of the database grid. Router will modify width to %.10g.	The Interactive Router cannot use the given width because it conflicts with the database grid, so the Interactive Router will adjust the width automatically to the given value.	The router will use a different width than what you specified. If this is not desired, compare the width against the database grid.
IGCR_CRE ATE_ROUTE E-024	Warning	Width %.10g is not an even multiple of the layer manufacturing grid %.10g. Router will modify width to %.10g.	The Interactive Router cannot use the given width because it conflicts with the layer's manufacturing grid, so the Interactive Router will adjust the width automatically to the given value.	The router will use a different width than what you specified. If this is not desired, compare the width against the layer's manufacturing grid and the top-level manufacturing grid.
IGCR_CRE ATE_ROUTE E-025	Warning	Width %f is not an even multiple of the manufacturing grid %f.	The Interactive Router cannot use the given width because it conflicts with the top-level manufacturing grid.	Set a width that is a multiple of the manufacturing grid.
IGCR_CRE ATE_ROUTE E-027	Warning	The layer %s has min %s constraint of %s	The routing parameters you have set might conflict with the specified constraint on the current layer.	Compare the parameters you have set with the constraint and make sure that this is the intended setting.
IGCR_CRE ATE_ROUTE E-067	Warning	There exists no valid routing LPP.	The Interactive Router did not find any valid layer-purpose pair to use for routing.	Check that the layer-purpose pairs are properly defined in the technology database and that a valid routing layer-purpose pair is selected and visible in the Object/Layer Panel.

ID	Type	Message	Causes	Solutions
IGCR_CRE ATE_ROUT E-070	Error	Cannot start irt::createRoute command. A severe problem was detected with the design constraints/data.\n Consult %s file.	The Interactive Router detected a severe error when attempting to load the data to start routing.	Examine the log file and try to fix the errors.
IGCR_CRE ATE_ROUT E-072	Error	Router has not been activated.	You are attempting to use a routing feature, but you have not yet enabled the router.	In the Layout Editor, choose Tools > Router , and then try starting the Interactive Router again.
IGCR_CRE ATE_ROUT E-077	Warning	Preference %s is deprecated and replaced by preference %s.	The cited Tcl preference has been deprecated.	Use the new Tcl preference.
IGCR_CRE ATE_ROUT E-078	Warning	Technology contained multiPatternSpacing rules. Preference rtFollowRoutingTracks is turned on during initialization to improve adherence to track/grid rules. Change the 'Follow Routing Tracks' option to 'off' if this setting is not desired.	Informational message only. On Track/Grid (rtFollowRoutingTracks) is turned on automatically when you enable the Interactive Router.	If you do not want routes to snap to tracks, you can turn off On Track/Grid in the Create Route Layer Tab.
IGCR_CRE ATE_ROUT E-082	Warning	Point (%d,%d) is outside of the defined routing area. Point ignored	You selected a route point outside of the defined routing area.	Select a point inside the area, or redefine the routing area to include the desired point, then try to route again. See Defining a Routing Area.
IGCR_CRE ATE_ROUT E-120	Warning	Could not instantiate Tcl script: %s	One of the script commands failed. The detailed message explains the reasons for failure. For example, you might have introduced an error in one of the Tcl scripts.	Contact Synopsys Support for scripting assistance.